

Definition-Consistent Variable-Order Fractional Operators from a Single Fixed-Pole Bank: Structure, Design Rule, and Fixed-Point Realization

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Abstract

Variable-order (VO) fractional operators have several inequivalent definitions that differ in how past orders enter the memory. When a finite-dimensional realization switches coefficients as the order varies, it is unclear which definition it implements. An exact switching state map exists only between similar realizations, so realizations whose poles move with the order cannot implement the A- or B-type Grünwald–Letnikov definitions under switching. A Beta-integral representation instead realizes the Grünwald–Letnikov weights with order-independent poles. With the order in the output weights the bank is the A-type, and with it in the input weights the B-type, for every order sequence; inverting it gives the recursive D- and E-types with exact duality, at $\mathcal{O}(K)$ cost per sample. In the ℓ^1 and ℓ^∞ operator norms the relative error of the forward types never exceeds the relative weight error; the recursive integrals amplify it by the condition number of the inverted derivative. The construction extends to every kernel family that is completely monotone in the lag, including time-varying distributed orders, with $K = \mathcal{O}(\ln(1/\varepsilon) \ln(R/\varepsilon))$ common poles for tolerance ε and memory R . A fit-free analytic rule sets the bank size and met all 400 specifications of a sealed holdout, and a DC floor makes every inverse stable. Moving-pole hot swapping stays at errors of 0.2 to 1 under switching at any size; residues fitted by sign-constrained linear programming save about a third of the poles. A bit-exact fixed-point core changes order and type per sample through a ROM address, and the bank streams multifractional Brownian motion.

Keywords: variable-order fractional calculus, Grünwald–Letnikov, diffusive representation, fixed-pole approximation, duality, fixed-point hardware, multifractional Brownian motion

1 Introduction

Fractional-order operators describe systems whose memory decays as a power law. When the decay exponent itself changes in time, the natural model is a variable-order (VO) operator, whose order $\alpha(t)$ is a function of time [1–4]. Unlike the constant-order case, the generalization is not unique. Different ways of letting the order enter the memory kernel give operators that coincide for constant order but differ as soon as the order changes [2, 5, 6].

For Grünwald–Letnikov (GL) operators, Sierociuk et al. use four types [7–10]:

- the A-type weights the whole history with the current order;
- the B-type weights each past sample with the order that was active when the sample entered;
- the recursive D- and E-types are the inverses of the A- and B-types of opposite order [8, 10, 11].

After a single order switch these definitions can differ by as much as the output itself (relative differences of 0.5 to 1.35 in Sect. 7). The choice of definition is therefore part of the model, not a detail of the numerics.

Practical realizations of fractional operators are finite-dimensional. Constant-order operators are approximated by rational transfer functions, most often by Oustaloup’s recursive distribution of poles and zeros [12], Charef’s fractional power pole [13] and related schemes [14]. The poles and zeros of these approximations move with the order. Fixed-pole approximations with order-independent poles have been proposed for constant orders [15–17]. Diffusive, infinite-state and sum-of-exponentials representations also lead to realizations with fixed nodes [18–22].

When the order varies, a constant-order realization can be switched between orders, either by switching circuits or by changing its coefficients while keeping its state (a “hot swap”). In our previous work on a constant-order FPGA core [23], such swaps produced short, bounded transients; which VO operator they implement was not addressed. Section 1.1 reviews how existing realizations handle a varying order.

This leaves two open questions:

1. Given an arbitrary order sequence, which VO definition does a finite-dimensional realization implement?
2. Can the structure of the realization select the definition at no extra cost per sample?

This paper answers both for GL operators of order $-1 < \alpha < 1$ and, more generally, for kernel families that are completely monotone in the lag. The contributions are:

1. *Necessity.* An exact switching state map exists only between similar realizations (Theorem 6). Realizations with order-dependent poles therefore cannot implement

the A- or B-type exactly at a switch, with or without a state map. Least-squares maps reduce the error in floating point. However, they need a dense $S \times S$ matrix per pair of orders, depend on the input class, and lose three to nine orders of magnitude of accuracy in 24-bit fixed point (Sect. 7.3).

2. *Definition by structure.* Order-independent exponents have already been used for the A-type VO Caputo derivative in fast numerical solvers [24, 25]. We show that, in a fixed-pole structure, the place of the order selects the definition. A Beta-integral representation of the GL weights gives a discrete bank with order-independent poles. With the order in the output weights the bank realizes the A-type, and with the order in the input weights the B-type. This holds for every order sequence, up to the frozen-order quadrature error (Proposition 2); the B-type thus costs $O(K)$ per sample as well. Algebraic inversion of the same bank gives the D- and E-types with exact duality at the same cost, and solves VO relaxation equations of all four types (Proposition 4). In analog form, a direct realization of the D-type has been considered impractical [10].
3. *Operator-level guarantees.* The weight error bounds the realized operators as a whole, for every order sequence (Theorem 5). In the ℓ^1 and ℓ^∞ operator norms the relative error of the A- and B-types never exceeds the relative weight error. The recursive integrals amplify it by the condition number of the inverted derivative, which grows like $N^{|\alpha|}$; one step response of the bank certifies their error a posteriori.
4. *Completely monotone families.* The structural results hold for every kernel family that is completely monotone in the lag, including time-varying distributed orders and fixed tempering. Under analyticity and power-law conditions on the density, a common pole set of size $K = O(\ln(1/\varepsilon) \ln(R/\varepsilon))$ meets a relative tolerance ε over a memory R uniformly in the order (Theorem 9).
5. *Fit-free design.* An analytic error model with three closed-form terms sets the bank size and pole range from the tolerance, the memory length and the order range. On a sealed holdout of 400 random specifications the rule met every tolerance (Sects. 5 and 7.4). Against moving-pole hot swapping, truncated convolution and fixed poles with residues fitted by linear programming, the fixed-pole structure is the only one that reaches small errors under switching; fitted residues reduce the bank size, but keep a stability certificate only with sign constraints (Sect. ??).
6. *Inverse stability.* The inverse of a derivative bank is stable if and only if its realized DC gain is positive (Proposition 8). This condition is tested through an exact sign rather than through eigenvalues. A DC floor applied through the delay mode guarantees it, also after coefficient quantization.
7. *Hardware.* A time-multiplexed fixed-point core realizes all four types. Order and type are per-sample inputs, and changing them only changes a ROM address. The core is bit-exact with its golden model (94 208 vectors, no mismatch). Word lengths for two operating envelopes and synthesis estimates are reported.
8. *Application.* The bank streams multifractional Brownian motion (mBm) of Riemann–Liouville type, which is an A-type integral of white noise, and its B-type counterpart, which is the memory-multifractional Brownian motion of [26] up to normalization, for $0 < H < 1$. One bank suffices because an integer split is exact when placed according to the type (Proposition 10).

Section 1.1 reviews related work. Section 2 recalls the definitions and their duality. Section 3 builds the continuous- and discrete-time fixed-pole banks, Section 4 proves the structural results and the operator-norm bounds, and Section 5 gives the error model, the design rule and the stability result. Section 6 extends the construction to completely monotone kernel families. Section 7 reports the numerical experiments, Section 8 the fixed-point core, and Section 9 the mBm application. Section 10 discusses limitations. All code, seeds and results are available (see the Declarations).

1.1 Related work

The search behind this section is documented in the repository: protocol, queries, screening decisions and data extraction. It used Crossref and arXiv and was complemented by snowballing. Table 1 summarizes how existing realizations treat a varying order.

Analog realizations.

Passive domino and nested ladders already show variable-order behaviour in the time domain [27]. Most analog VO realizations, however, switch between two or three constant-order circuits [28, 29], as noted in [30, 31]. Sierociuk et al. made the link between structure and definition explicit for switching:

- a serial cascade in which a block of complementary order is pre-connected at every switch is equivalent to the B-type [7];
- the recursive types are dual to the iterative ones [8, 32], and switching strategies for them follow from this duality [8, 9, 33];
- a parallel scheme replaced the serial one, which could not be built in practice [34].

The D-type was found impractical to realize directly, because two ideal operators of opposite orders cannot be connected in series with electronic elements [10]. The serial scheme adds a block at every switch [7], and the analog models built so far cover a few switches. Arıcioglu realized a smoothly varying order with a single transfer function [30]. This restricts the method to the only linear time-invariant definition of Lorenzo and Hartley [2]. Electronically tunable fractional-order elements and filters [35, 36] and variable fractional-order digital differentiators and integrators [37–39] make the order an adjustable parameter of a constant-order design. Yu et al. went further and changed the element values of a fractal-ladder fractor with programmable resistors and capacitors, both in jumps and continuously [31]. They compared the response with a GL reference whose weights take the order at the time of each sample, i.e. a B-type reference. Because the element values, and hence the poles, change with the order, Theorem 6 below implies that such a match can only be approximate.

Digital and hardware realizations.

Tolba et al. realized GL operators on an FPGA with an order that can be reconfigured at run time, and used them in a VO chaotic oscillator [40]. A direct GL convolution whose weights are recomputed for the current order has fixed poles (all at the origin) and its order in the output weights. By the argument of Proposition 2 below it is then an A-type operator, with a memory and a cost per sample proportional to its length.

The discrete-time VO PID controller of Oziablo et al. evaluates exactly such an A-type GL difference [41]. Its cost per step grows with time, and the authors note that bounding the stored history without disturbing the control is an open problem.

Fixed nodes for variable orders.

The work closest to ours comes from fast solvers for VO time-fractional equations. Zhang, Fang and Sun approximate the VO Caputo derivative, with the order taken at the current time, by an exponential sum whose exponents do not change between time levels [24, 25]. Huang et al. propose sum-of-exponentials approximations for variable-order kernels within a unified fast time-stepping framework [42]. The exponential-sum approximation of Zhang et al. shares the order-independent nodes with the present paper. However, it treats only the A-type, in continuous time and inside an L1-type discretization. It does not address the B-, D- or E-types, a design rule over the whole order range, or hardware. For the hidden-memory VO derivative, whose order is evaluated at the integration variable as in the B-type, the fast method we found uses Taylor expansions and hierarchical matrices at $O(N \log N)$ total cost [43].

Time-varying fractional filters.

In time-series analysis, Philippe, Surgailis and Viano introduced two time-varying fractional filters $A(d)$ and $B(d)$ built from a sequence d_t [44, 45]. Both reduce to $(I - L)^{-d}$ for constant d and satisfy $B(-d)A(d) = A(-d)B(d) = I$. Their coefficients are products of d_u over the times between input and output, so they differ from the GL types used here. Their exact inverse relation is, however, an analogue of the duality (7). Surgailis extended this construction to continuous time to define multifractional processes [46].

State and transients.

Transients caused by coefficient changes in recursive filters are a classical topic [47, 48]. The meaning of the state of a fractional system has been discussed in the context of initialization [19, 49, 50]. Neither line asks which VO definition a switched realization implements.

2 Variable-order Grünwald–Letnikov operators

2.1 Definitions

Let T_s be the sampling period and $\alpha_n \in (-1, 1)$ the order at sample $n \geq 0$. An order $\alpha > 0$ is a derivative, $\alpha < 0$ an integral and $\alpha = 0$ the identity. Signals vanish for $n < 0$. We write

$$w_r(a) = (-1)^r \binom{a}{r} = w_{r-1}(a) \frac{r-1-a}{r}, \quad w_0 = 1, \quad g_r(a) = T_s^{-a} w_r(a). \quad (1)$$

Table 1 How existing realizations treat a varying order. “Inferred” marks entries obtained by applying Proposition 2 or Theorem 6 to the structure described in the cited work

Work	Structure	Definition	Order changes	Cost per sample
serial switching [7]	analog; complementary-order block added at each switch	B	any number of switches	grows with switches
recursive, parallel schemes [8, 10, 34] [30]	analog; switched constant-order ladders analog; one transfer function	D; B; B $\leftrightarrow$ D LTI (third) type of [2]	a few switches smooth $\alpha(t)$	— fixed
tunable elements, filters [35–37, 39]	order as a design parameter	FIR with current- order coefficients: A (inferred)	re-tuning	—
programmable ladder [31]	element values change with the order (moving poles)	compared with B; approximate (Thm. 6)	jumps, continuous	fixed
FPGA GL [40]; VO PID [41]	direct GL convolution	A (inferred [40]; stated [41])	at run time	$O(L)$, $L =$ memory
hot-swapped Oustaloup (Sect. 7)	moving poles, state kept	none exactly (Thm. 6)	any	$O(S)$
exponential sums [24, 25]	fixed exponents, continuous-time Caputo	A	any $\alpha(t) \in (0, 1)$	$O(\log^2 n)$
hidden memory [43]	Taylor / hierarchical matrices	B-like (Caputo)	any	$O(N \log N)$ in total
this work	discrete fixed-pole bank	A, B; D, E by inversion	every sample	$O(K)$, $K \approx 20\text{--}50$

Definition 1 (VO GL operators [7, 8, 10]) For an input x and an order sequence α ,

$$({}^A\Delta^\alpha x)_n = \sum_{r=0}^n g_r(\alpha_n) x_{n-r}, \quad (2)$$

$$({}^B\Delta^\alpha x)_n = \sum_{r=0}^n g_r(\alpha_{n-r}) x_{n-r}, \quad (3)$$

$$z_n = ({}^D\Delta^\alpha x)_n = T_s^{-\alpha_n} x_n - \sum_{r=1}^n w_r(-\alpha_n) z_{n-r}, \quad (4)$$

$$z_n = ({}^E\Delta^\alpha x)_n = T_s^{-\alpha_n} x_n - \sum_{r=1}^n w_r(-\alpha_{n-r}) T_s^{\alpha_{n-r}-\alpha_n} z_{n-r}. \quad (5)$$

The A-type has no memory of past orders: a change of α_n immediately reweights the entire history. The B-type keeps each sample with the weight sequence of its own entry order. For a constant order all four types coincide. In [7] the A- and B-types are the first and second types. Their third type indexes the order by the lag, which gives a time-invariant operator. Lorenzo and Hartley [2] and Valério and Sá da Costa [5] list further variants, and [51] exploits the duality below to solve VO equations analytically. The time-varying fractional filters of [44, 45] form a different family with an exact inverse relation of the same kind.

2.2 Matrix form and duality

On a horizon of N samples the A- and B-types are lower-triangular matrices,

$$[W_A(\alpha)]_{n,j} = g_{n-j}(\alpha_n), \quad [W_B(\alpha)]_{n,j} = g_{n-j}(\alpha_j), \quad 0 \leq j \leq n < N. \quad (6)$$

Row n of $W_A(-\alpha)z = x$ reads $T_s^{\alpha_n} z_n + T_s^{\alpha_n} \sum_{r \geq 1} w_r(-\alpha_n) z_{n-r} = x_n$, which is (4). Row n of $W_B(-\alpha)z = x$ is (5). Hence

$$W_D(\alpha) = W_A(-\alpha)^{-1}, \quad W_E(\alpha) = W_B(-\alpha)^{-1}, \quad (7)$$

which is the duality of [8, 10, 11]. For a varying order, the A-types (or the B-types) of opposite orders are not inverses of each other: $W_A(\alpha)W_A(-\alpha) \neq I$ in general. The recursive types are therefore genuinely different operators. Section 7.5 quantifies these relations numerically.

2.3 Continuous-time references

For the continuous-time experiments we use the step responses of the A- and B-types. With $F(t, a) = t^{-a}/\Gamma(1-a)$, the step response of the constant-order operator s^a , they are

$$y_A(t) = F(t, \alpha(t)), \quad (8)$$

$$y_B(t) = F(t, \alpha(0)) + \int_0^t \partial_a F(t - \tau, \alpha(\tau)) \dot{\alpha}(\tau) d\tau. \quad (9)$$

Equation (9) is derived in Appendix C. For a piecewise-constant order with switches at T_i , (9) reduces to $F(t, \alpha_0) + \sum_i [F(t - T_i, \alpha_i) - F(t - T_i, \alpha_{i-1})]$.

2.4 Diffusive representation

For $0 < a < 1$ and $S_a = \sin(\pi a)/\pi$ [18, 22],

$$s^{-a} = S_a \int_0^\infty \frac{\xi^{-a}}{s + \xi} d\xi, \quad s^a = S_a \int_0^\infty \xi^{a-1} \frac{s}{s + \xi} d\xi. \quad (10)$$

The order enters only through the density in ξ ; the pole $-\xi$ of each element does not depend on it. This is the property the paper builds on.

3 Fixed-pole banks

3.1 Continuous time

Let $\xi_k = \xi_0 e^{kh}$, $k = 0, \dots, K-1$, be a geometric grid with spacing h in $\ln \xi$, and apply the trapezoidal rule to (10). Every order is then written on the same bank of

unit-DC-gain states $\dot{x}_k = -\xi_k x_k + \xi_k u$:

$$H_\alpha(s) = d(\alpha) + \sum_{k=0}^{K-1} m_k(\alpha) \frac{\xi_k}{s + \xi_k}. \quad (11)$$

For an integral, $\alpha = -a < 0$, the weights are $m_k = h S_a \xi_k^{-a}$. For a derivative, $\alpha = a > 0$, we use $s/(s + \xi) = 1 - \xi/(s + \xi)$: then $m_k = -v_k$ with $v_k = h S_a \xi_k^a$, and the feedthrough collects $\sum_k v_k$. The two truncated tails are closed by the geometric series of the virtual trapezoid nodes beyond the grid:

- one tail is lumped onto the outermost pole: $m_0 += S_a \xi_{\text{lo}}^{-a} G_h(1 - a)$ for integrals, $v_{K-1} += S_a \xi_{\text{hi}}^a G_h(1 - a)$ for derivatives;
- the other becomes feedthrough: $d = S_a \xi_{\text{hi}}^{-a} G_h(a)$ for integrals, $d = \sum_k v_k + S_a \xi_{\text{lo}}^a G_h(a)$ for derivatives.

Here

$$G_h(p) = \sum_{j \geq 1} h e^{-pjh} = \frac{h}{e^{ph} - 1}, \quad H_h(p, q) = G_h(p) - G_h(q). \quad (12)$$

All nodes carry the full weight h . The bank therefore equals the infinite trapezoidal rule except for asymptotic tail residuals, and no $O(h^2)$ endpoint term appears. At $\alpha = 0$ the bank is the identity ($m = 0, d = 1$). Since $S_a G_h(a) \rightarrow 1$ as $a \rightarrow 0$, the coefficients are continuous when the order crosses zero.

3.2 Discrete time: a Beta-integral representation of the GL weights

Sampling the continuous-time bank introduces a second error, because modes faster than $1/T_s$ respond to the staircase of a sampled order (Sect. 7.2). We therefore realize the GL weights (1) themselves.

Lemma 1 For $r \geq 1$ and $-1 < a < 1$,

$$w_r(a) = -S_a B(r - a, 1 + a) = -S_a \int_0^1 \theta^{r-a-1} (1 - \theta)^a d\theta. \quad (13)$$

Proof $w_r(a) = \Gamma(r - a)/(\Gamma(-a)\Gamma(r + 1))$ and $B(r - a, 1 + a) = \Gamma(r - a)\Gamma(1 + a)/\Gamma(r + 1)$. The reflection formula gives $\Gamma(-a)\Gamma(1 + a) = -\pi/\sin(\pi a)$. For $a = 0$ both sides vanish. $\square$

The weights are thus a superposition of geometric sequences θ^r with an order-dependent density. With $\theta = e^{-u}$ and $u = e^x$,

$$w_r(a) = -S_a \int_{-\infty}^{\infty} \varphi_a(e^x) e^{-(r-1)e^x} dx, \quad \varphi_a(u) = u e^{-(1-a)u} (1 - e^{-u})^a. \quad (14)$$

The trapezoidal rule on $u_k = u_{\text{lo}} e^{kh}$, $k = 0, \dots, K-1$, gives order-independent discrete poles $\theta_k = e^{-u_k}$. With $u_k = \xi_k T_s$, each θ_k is the zero-order-hold image of the continuous-time pole $-\xi_k$. The realized weights are

$$\begin{aligned} \hat{g}_0(a) &= T_s^{-a}, & \hat{g}_r(a) &= \delta_{r1} c_d(a) + \sum_{k=0}^{K-1} c_k(a) \theta_k^{r-1} \quad (r \geq 1), \\ c_k(a) &= -S_a T_s^{-a} h \varphi_a(u_k), \end{aligned} \quad (15)$$

and the frozen-order transfer function is

$$\hat{H}_a(z) = \sum_{r \geq 0} \hat{g}_r(a) z^{-r} = T_s^{-a} + \frac{c_d(a)}{z} + \sum_{k=0}^{K-1} \frac{c_k(a)}{z - \theta_k}. \quad (16)$$

The tails are again closed by virtual trapezoid nodes:

- the nodes below the grid ($\theta \approx 1$) are lumped onto θ_0 , matched at $r = 1$: $c_0 += -S_a T_s^{-a} \sum_{j \geq 1} h \varphi_a(u_{\text{lo}} e^{-jh})$;
- the nodes above the grid ($\theta \approx 0$) contribute only at $r = 1$ and form the *delay mode* (a pole at $\theta = 0$): $c_d = -S_a T_s^{-a} \sum_{j \geq 1} h \varphi_a(u_{\text{hi}} e^{jh})$.

Since $\varphi_a > 0$, all residues have the sign of $-a$: integral banks have positive residues and derivative banks negative ones (before the DC floor of Sect. 5.4). The weight $\hat{g}_0$ is exact, and at $a = 0$ the bank is exactly the identity. As $a \rightarrow 0$ all residues vanish continuously, so the bank crosses the order zero without any special case.

3.3 Order schedules

The bank has K first-order states s_k and the delay state s_d . The order can be scheduled in two places (Fig. 1), the output schedule (OS) and the input schedule (IS):

$$\begin{aligned} \text{OS:} \quad & s_k[n+1] = \theta_k s_k[n] + v_n, & s_d[n+1] &= v_n, \\ & y_n = T_s^{-\alpha_n} v_n + \underbrace{c_d(\alpha_n) s_d[n] + \sum_k c_k(\alpha_n) s_k[n]}_{\eta_n}, \end{aligned} \quad (17)$$

$$\begin{aligned} \text{IS:} \quad & s_k[n+1] = \theta_k s_k[n] + c_k(\alpha_n) v_n, & s_d[n+1] &= c_d(\alpha_n) v_n, \\ & y_n = T_s^{-\alpha_n} v_n + \underbrace{s_d[n] + \sum_k s_k[n]}_{\eta_n}. \end{aligned} \quad (18)$$

In both cases $y_n = \hat{g}_0(\alpha_n) v_n + \eta_n$, where the history η_n depends only on $v_0, \dots, v_{n-1}$ because the poles do not depend on the order. Changing the order never touches the state.

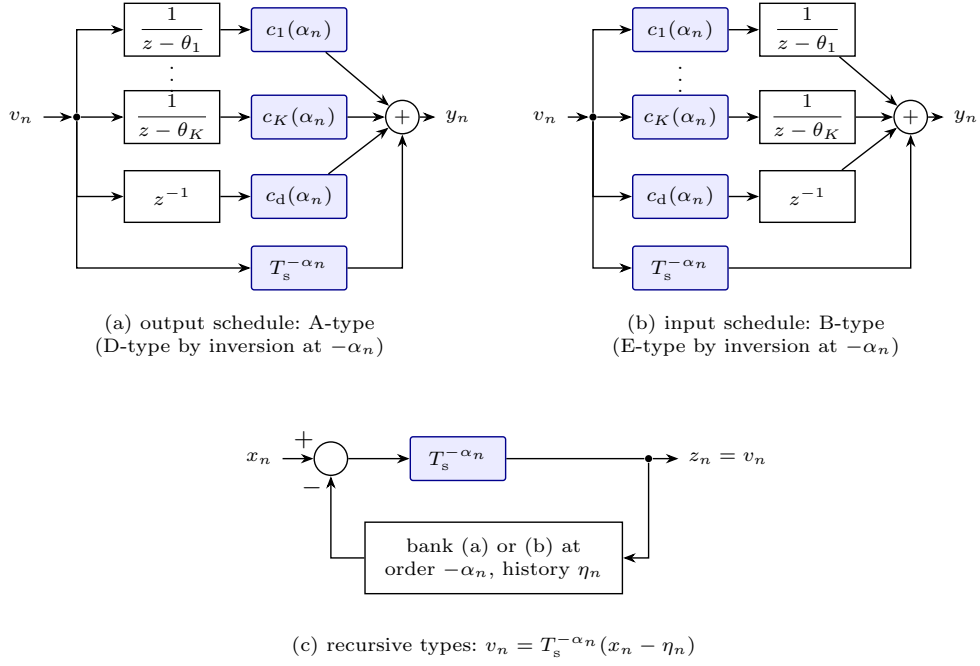


Fig. 1 Fixed-pole bank. The poles θ_k and the delay z^{-1} are common to every order; only the shaded weights depend on α_n . (a) Order in the output weights. (b) Order in the input weights. (c) The recursive types: the history η_n of bank (a) or (b), run at order $-\alpha_n$, is removed and the result is scaled by $1/\hat{g}_0(-\alpha_n) = T_s^{-\alpha_n}$. The feedthrough of the bank at order $-\alpha_n$ is $T_s^{\alpha_n}$, which is why the gain in (c) is $T_s^{-\alpha_n}$.

4 Definition by structure

Proposition 2 (Definition by structure) *For every order sequence $(\alpha_n) \subset (-1, 1)$, every input v and zero initial state,*

$$OS: y_n = \sum_{r=0}^n \hat{g}_r(\alpha_n) v_{n-r}, \quad IS: y_n = \sum_{r=0}^n \hat{g}_r(\alpha_{n-r}) v_{n-r}. \quad (19)$$

The output-scheduled bank is therefore the A-type (2), and the input-scheduled bank the B-type (3), with the realized weights $\hat{g}_r$ in place of g_r .

Proof In OS, $s_k[n] = \sum_{r=1}^n \theta_k^{r-1} v_{n-r}$ and $s_d[n] = v_{n-1}$. Hence $\eta_n = \sum_{r \geq 1} (\delta_{r1} c_d(\alpha_n) + \sum_k c_k(\alpha_n) \theta_k^{r-1}) v_{n-r} = \sum_{r \geq 1} \hat{g}_r(\alpha_n) v_{n-r}$. In IS, $s_k[n] = \sum_{r=1}^n \theta_k^{r-1} c_k(\alpha_{n-r}) v_{n-r}$ and $s_d[n] = c_d(\alpha_{n-1}) v_{n-1}$, so $\eta_n = \sum_{r \geq 1} \hat{g}_r(\alpha_{n-r}) v_{n-r}$. $\square$

The state is never modified when the order changes, and no switching term appears. Accuracy is set by the frozen-order design alone:

Corollary 3 Let $\varepsilon_R = \sup_a \max_{1 \leq r < R} |\hat{g}_r(a)/g_r(a) - 1|$ over the design range of orders. For $n < R$,

$$\begin{aligned} \text{OS: } \quad & \left| y_n - (\mathbf{A}\Delta^\alpha v)_n \right| \leq \varepsilon_R \sum_{r=1}^n |g_r(\alpha_n)| |v_{n-r}|, \\ \text{IS: } \quad & \left| y_n - (\mathbf{B}\Delta^\alpha v)_n \right| \leq \varepsilon_R \sum_{r=1}^n |g_r(\alpha_{n-r})| |v_{n-r}|. \end{aligned} \quad (20)$$

The bound is relative to the weights, not to the output. For derivatives the weights nearly cancel ($\sum_r g_r(a) = 0$ for $a > 0$). Relative output errors for smooth inputs can therefore exceed ε_R (Sect. 7.5).

Proposition 4 (Recursive types by inversion) Run the OS (IS) bank at the order sequence $-\alpha$ and let η_n be its history. The causal recursion

$$v_n = \frac{x_n - \eta_n}{\hat{g}_0(-\alpha_n)} = T_s^{-\alpha_n}(x_n - \eta_n), \quad z_n = v_n, \quad (21)$$

gives $z = \hat{W}_A(-\alpha)^{-1}x$ (respectively $\hat{W}_B(-\alpha)^{-1}x$), where $\hat{W}$ is (6) with the realized weights. It thus realizes the D-type (E-type) with realized weights, and the duality (7) holds exactly in exact arithmetic:

$$\hat{\mathbf{A}}\Delta^{-\alpha} \hat{\mathbf{D}}\Delta^\alpha = \hat{\mathbf{D}}\Delta^\alpha \hat{\mathbf{A}}\Delta^{-\alpha} = I, \quad \hat{\mathbf{B}}\Delta^{-\alpha} \hat{\mathbf{E}}\Delta^\alpha = \hat{\mathbf{E}}\Delta^\alpha \hat{\mathbf{B}}\Delta^{-\alpha} = I.$$

Each sample costs $O(K)$, the same as the forward types.

Proof By Proposition 2, row n of $\hat{W}_A(-\alpha)v = x$ is $\hat{g}_0(-\alpha_n)v_n + \eta_n = x_n$. Since $\hat{g}_0(-\alpha_n) = T_s^{\alpha_n} \neq 0$, the triangular system is solved forward by (21). The IS case is identical. $\square$

Remark 1 (VO relaxation equations) The same structure solves ${}^T\Delta^\alpha y + \lambda y = u$ for $T \in \{A, B, D, E\}$ at $O(K)$ per sample:

- A and B: $y_n = (u_n - \eta_n)/(\hat{g}_0(\alpha_n) + \lambda)$;
- D and E: write $y = \hat{W}(-\alpha)z$ with $z = u - \lambda y$, which gives $y_n = (\hat{g}_0(-\alpha_n)u_n + \eta_n)/(1 + \lambda\hat{g}_0(-\alpha_n))$, the bank being driven by $z_n = u_n - \lambda y_n$.

A direct triangular solve of the GL equations costs $O(N^2)$.

Corollary 3 bounds each output sample by a weighted sum of past inputs. Because every row of the A-type error matrix and every column of the B-type error matrix holds a single order, the same weight errors also bound the realized operators as a whole, for every order sequence. Let $\|\cdot\|_p$ denote the induced ℓ^p norm of an $N \times N$ matrix and $|M|$ the entrywise absolute value.

Theorem 5 (Operator-norm bounds) Let $\mathcal{S} \subset (-1, 1)$ be a set of orders, $N \leq R$, and $e_r(a) = \hat{g}_r(a) - g_r(a)$ for $1 \leq r < N$. Put

$$\rho_{\text{row}} = \sup_{a \in \mathcal{S}} \sum_{r=1}^{N-1} |e_r(a)|, \quad \rho_{\text{col}} = \sum_{r=1}^{N-1} \sup_{a \in \mathcal{S}} |e_r(a)|. \quad (22)$$

For every order sequence with values in $\mathcal{S}$, let $E_T = \hat{W}_T - W_T$ be the error of the realized operator of type T .

- (a) $\|E_A\|_\infty \leq \rho_{\text{row}}$, $\|E_A\|_1 \leq \rho_{\text{col}}$, $\|E_B\|_1 \leq \rho_{\text{row}}$, $\|E_B\|_\infty \leq \rho_{\text{col}}$, and $\|E_T\|_2 \leq (\rho_{\text{row}}\rho_{\text{col}})^{1/2}$ for $T \in \{A, B\}$. If $\mathcal{S}$ is finite, each of the four bounds is attained by some order sequence.
- (b) $\|E_T\|_p \leq \varepsilon_R \|W_T\|_p$ for $1 \leq p \leq \infty$ and $T \in \{A, B\}$. In particular, $\|E_T\|_p \leq \varepsilon_R \|W_T\|_p$ for $p \in \{1, \infty\}$: the relative error of the operator, and hence the worst-case relative output error over all inputs, never exceeds the relative weight error.
- (c) For $(T, F) \in \{(D, A), (E, B)\}$, let E_F^- be the error of $\hat{W}_F(-\alpha)$ and $q = \|W_T\|_p \|E_F^-\|_p$. If $q < 1$, then $\|E_T\|_p \leq \frac{q}{1-q} \|W_T\|_p$. The same holds with $\hat{q} = \|\hat{W}_T\|_p \|E_F^-\|_p$ and $\|\hat{W}_T\|_p$ in place of q and $\|W_T\|_p$. If all orders are negative and $\hat{g}_1(-\alpha_n) < 0$ for all n , then $\hat{W}_T \geq 0$ entrywise, and $\|\hat{W}_T\|_\infty$ is the peak of the realized step response.

The proof is given in Appendix E. Part (a) uses the error kernel itself and does not depend on how the order sequence is chosen. Part (c) gives an a-posteriori certificate for the recursive integrals: one step response of the bank ($O(NK)$) and the constants (22) at the opposite orders.

Remark 2 (Conditioning of the recursive types) By (b), $q \leq \varepsilon_R \kappa_p$ with $\kappa_p = \|W_F(-\alpha)\|_p \|W_F(-\alpha)^{-1}\|_p$. For a constant order $a \in (0, 1)$, the partial sums of $(1-z)^{\pm a}$ give

$$\kappa_\infty = (2 - w_{N-1}(a-1)) w_{N-1}(-a-1) \approx \frac{2N^a}{\Gamma(1+a)},$$

which is 71 for $a = 0.5$ and $N = 1000$. For a recursive integral this amplification is real. Its step response is set by the DC gain of the inverted derivative bank, which is small (about $T_s^{-a} N^{-a} / \Gamma(1-a)$) and carries an absolute error of order $\varepsilon_R T_s^{-a}$. A recursive integral with relative operator error ε therefore needs a bank designed for about $\varepsilon / \kappa_\infty$. For recursive derivatives the bound is conservative; the measured errors stay at the level of ε_R (Sect. 7.6).

The fixed-pole structure is not only sufficient for definition consistency. Some version of it is also necessary:

Theorem 6 (Necessity) *Let $\Sigma_i = (F_i, b_i, c_i^\top, d_i)$, $i = 1, 2$, be minimal realizations of the same dimension S of two frozen-order operators. The switched system runs Σ_1 for $n < m$ and Σ_2 for $n \geq m$, where $m \geq S+1$, and the state is mapped as $x[m] \leftarrow \Phi x[m]$ at the switch.*

- (a) *For every input, the output for $n \geq m$ equals the response of Σ_2 to the entire input (A-type behaviour) if and only if $F_2 = \Phi F_1 \Phi^{-1}$ and $b_2 = \Phi b_1$.*
- (b) *For every input, the output for $n \geq m$ equals the response of Σ_1 to the inputs before m plus the response of Σ_2 to the inputs from m on (B-type behaviour) if and only if $F_2 = \Phi F_1 \Phi^{-1}$ and $c_2^\top = c_1^\top \Phi^{-1}$.*

In both cases F_1 and F_2 are similar, so the two realizations have the same poles.

The proof is given in Appendix A. The same statement holds in continuous time with $e^{F_i t}$ in place of F_i^n .

Corollary 7 *If the poles of a frozen-order realization depend on the order, as in the Oustaloup family [12], then no state map realizes the A- or B-type exactly at a switch between orders with different poles. In particular the hot swap $\Phi = I$ does not. The fixed-pole bank satisfies (a) with $\Phi = I$ in OS, where (F, b) does not depend on the order, and (b) with $\Phi = I$ in IS, where $(F, c^\top)$ does not depend on the order.*

Remark 3 (Approximate maps) Theorem 6 rules out exact maps, not approximate ones. A least-squares Φ fitted to an input ensemble can make the definitional error small in floating point (Sect. 7.3), because the reachable states of both realizations concentrate in a low-dimensional subspace. Such a map, however:

- is dense, costing S^2 multiply-accumulates (MACs) at every switch;
- is specific to each pair of orders, needing $2(P-1)$ maps for the $P-1$ neighbouring pairs of a grid of P order levels;
- depends on the statistics of the input.

State-update strategies for time-varying recursive filters suppress the transients of a coefficient change [47, 48]. By Theorem 6, no such update can make a moving-pole realization definition-consistent.

5 Error model, design rule and inverse stability

The design rule chooses h , u_{lo} , u_{hi} and hence K from a tolerance ε , a memory length R (in samples) and an order range $[\alpha_{\min}, \alpha_{\max}]$. In discrete time the metric is the relative weight error ε_R of Corollary 3. In continuous time it is $\sup_{\omega_1 \leq \omega \leq \omega_2} |H_\alpha(j\omega)/(j\omega)^\alpha - 1|$. Every term below follows from the representation, and no constant is fitted.

5.1 Discrete time

With the tail closure, the realized weights are the infinite trapezoidal rule applied to (14), apart from two tail residuals. The error therefore has three terms.

Quadrature.

The integrand of (14) is analytic in a strip around the real axis, so the error of the infinite trapezoidal rule is the aliasing of its Fourier transform at $2\pi/h$ (Poisson summation) [52]. For large r the integrand is $e^{(1+a)x}e^{-(r-a)e^x}$ to leading order. The modulus of its transform at $2\pi/h$ is $|\Gamma(1+a-2\pi i/h)|$ relative to $\Gamma(1+a)$, and Stirling's formula gives (Appendix D)

$$E_q^\infty(a, h) = \frac{2\sqrt{2\pi} (2\pi/h)^{a+1/2} e^{-\pi^2/h}}{\Gamma(1+a)}. \quad (23)$$

For small r the alias amplitude is evaluated directly from the representation. The relative errors of the infinite trapezoidal sum at the grid offsets 0 and $h/4$ are the cosine

Table 2 Continuous-time error terms (relative frequency-response error on $\omega_1 \leq \omega \leq \omega_2$), with $S = \sin(\pi a)/\pi$ and G_h, H_h from (12)

Term	integral, $\alpha = -a < 0$	derivative, $\alpha = a > 0$
quadrature (first alias)	$2 \sin(\pi a) e^{-\pi^2/h}$	$2 \sin(\pi a) e^{-\pi^2/h}$
lower tail	$S(\xi_{\text{lo}}/\omega_1)^{2-a} H_h(1-a, 2-a)$	$S(\xi_{\text{lo}}/\omega_1)^{1+a} G_h(1+a)$
upper tail	$S(\omega_2/\xi_{\text{hi}})^{1+a} G_h(1+a)$	$S(\omega_2/\xi_{\text{hi}})^{2-a} H_h(1-a, 2-a)$

and sine components of the first alias, and their root sum of squares is its amplitude. We use $E_q = \max(E_q^\infty, \max_{r \in \{1, 2, 3, 5, 8\}} E_q^{(r)})$.

Lower tail.

The virtual nodes below u_{lo} are matched to θ_0 at $r = 1$. At lag r their mismatch is $\sum_j h \varphi_a(u_{-j})(\theta_{-j}^{r-1} - \theta_0^{r-1}) \approx (r-1)u_{\text{lo}}^{2+a} H_h(1+a, 2+a)$ times $-S_a T_s^{-a}$. Relative to $|w_r(a)| = |S_a|B(r-a, 1+a)$ this gives the residual

$$E_{\text{lo}} = \frac{(r-1)u_{\text{lo}}^{2+a} H_h(1+a, 2+a)}{B(r-a, 1+a)}, \quad \text{worst at } r = R-1. \quad (24)$$

It grows with the lag, so it is the term that ties u_{lo} to the memory length.

Upper tail.

The delay mode matches the nodes above u_{hi} at $r = 1$, so the first residual appears at $r = 2$:

$$E_{\text{hi}} = \frac{1}{B(2-a, 1+a)} \sum_{j \geq 1} h u_j e^{-(2-a)u_j} (1 - e^{-u_j})^a, \quad u_j = u_{\text{hi}} e^{jh}. \quad (25)$$

5.2 Continuous time

The same reasoning applied to (10) gives the terms of Table 2. The quadrature term is $2|\sin \pi a|e^{-\pi^2/h}$. The lower and upper tails exchange their forms between integrals and derivatives, as a consequence of the substitution $\xi \rightarrow 1/\xi$ in (10).

5.3 Rule

The tolerance is split as $\varepsilon/2$ for the quadrature term and $\varepsilon/4$ for each tail. Over a grid of orders covering the design range, the rule proceeds in four steps:

1. Choose the largest h with $\max_a E_q(a, h) \leq \varepsilon/2$. In continuous time this is the closed form $h = \pi^2 / \ln(4 \max_a |\sin \pi a|/\varepsilon)$; in discrete time it is a one-dimensional root.
2. For each order, solve the lower-tail condition in closed form, since it is a power law in u_{lo} (in ξ_{lo}/ω_1 in continuous time), and take the smallest value.
3. For each order, find u_{hi} (or ξ_{hi}) by a monotone root and take the largest value.
4. Set $K = \lceil \ln(u_{\text{hi}}/u_{\text{lo}})/h \rceil + 1$; the realized spacing then does not exceed the h of step 1.

5.4 Inverse stability and the DC floor

The recursive types (21) run the frozen-order inverse $1/\hat{H}$. Its dynamics matrix is $F - bc^\top/\hat{g}_0$ with $F = \text{diag}(\theta_0, \dots, \theta_{K-1}, 0)$, $b = \mathbf{1}$ and $c = (c_0, \dots, c_{K-1}, c_d)$. For the input schedule the dynamics matrix is $F - cb^\top/\hat{g}_0$, the transpose structure. Both have the characteristic polynomial $N(z)/\hat{g}_0$ with

$$N(z) = z \prod_k (z - \theta_k) \hat{H}(z).$$

The inverse is therefore stable if and only if all roots of N lie in the open unit disc.

Proposition 8 (Inverse stability) *Let $\hat{H}(z) = g_0 + c_d/z + \sum_k c_k/(z - \theta_k)$ with $g_0 > 0$ and distinct $\theta_k \in (0, 1)$.*

- (i) *If $c_k > 0$ for all k and $c_d \geq 0$ (integral bank), all roots of N are real and $1/\hat{H}$ is stable if and only if $\hat{H}(-1) > 0$.*
- (ii) *If $c_k < 0$ for all k (derivative bank), all roots of N are real and $1/\hat{H}$ is stable if and only if $\hat{H}(1) > 0$ and either $c_d \leq 0$ or $\hat{H}(-1) > 0$.*
- (iii) *If $|\hat{g}_r/g_r - 1| \leq 1/3$ for $r = 1, 2$, then $\hat{H}(-1) > 0$.*

Hence, for any bank whose first two weights are accurate to $1/3$ (which every design of the rule satisfies by a wide margin), an integral bank always has a stable inverse, and a derivative bank has one if and only if its realized DC gain $\hat{H}(1) = \sum_r \hat{g}_r$ is positive.

The proof (Appendix B) has two ingredients:

- interlacing: between consecutive poles with residues of equal sign there is exactly one root;
- monotonicity of $z\hat{H}(z)$ on $z > \max_k \theta_k$ for a derivative bank, which locates the root closest to $z = 1$.

The exact value of $\hat{H}(-1)$ is $\sum_r g_r(-1)^r = 2^a T_s^{-a}$. On the 25 orders checked in each of the 200 discrete-time holdout designs of Sect. 7.4, with and without the floor, the realized value is within 0.06 % of it.

For an exact GL derivative, $\sum_r g_r(a) = 0$, so the inverse has a pole on the unit circle in the limit. The finite memory of the bank adds a small positive DC term, approximately

$$S_a T_s^{-a} u_{lo}^a / (a(1+a)),$$

the part of the density below u_{lo} not represented by the lumped closure. For large orders and small u_{lo} this term is smaller than the quadrature error of the DC sum, and $\hat{H}(1)$ can become negative. We therefore apply a DC floor: if $\hat{H}(1) < H_{\min} = 0.01 |S_a| T_s^{-a} u_{lo}^a / (a(1+a))$, the delay residue is raised by $H_{\min} - \hat{H}(1)$. This changes only $\hat{g}_1$, by about the DC quadrature error, and makes every inverse stable by Proposition 8.

Remark 4 (Signs instead of eigenvalues) The dominant pole of a derivative-bank inverse lies extremely close to $z = 1$, and general eigenvalue solvers are unreliable there. In one holdout design (order 0.95, $K = 35$), a 50-digit bisection places the dominant pole at $1 - 2.9 \times 10^{-12}$, while a double-precision eigenvalue routine reports modulus $1 + 4.5 \times 10^{-9}$. The signs of $\hat{H}(1)$ and $\hat{H}(-1)$, in contrast, are computed to full relative precision, and in fixed point they are evaluated exactly (Sect. 8.3).

6 Beyond Grünwald–Letnikov: completely monotone kernel families

Sections 3–5 used two properties of the GL weights. For each frozen order they are a superposition of geometric sequences with a density of one sign, and this density is smooth on a logarithmic scale. Both properties hold for a much larger class of kernels.

Definition 2 (CM family) A family of causal kernels $\{k_a\}_{a \in \mathcal{A}}$ is a *completely monotone (CM) family* if $k_a(0) \neq 0$ and, for $r \geq 1$,

$$k_a(r) = \sigma_a \int_0^\infty e^{-(r-1)u} m_a(u) du, \quad \sigma_a \in \mathbb{R}, \quad m_a \geq 0. \quad (26)$$

Its A- and B-type operators are (2) and (3) with $k_a(r)$ in place of $g_r(a)$, and its D- and E-types are defined by the duality (7).

With $\theta = e^{-u}$, (26) states that $(k_a(r)/\sigma_a)_{r \geq 1}$ is the moment sequence of a nonnegative measure on $(0, 1)$. By Hausdorff’s theorem these are exactly the completely monotone sequences [53]; here the measure has a density, and an atom at $\theta = 0$ is the delay mode. For the GL weights, Lemma 1 gives $\sigma_a = -S_a T_s^{-a}$ and $m_a(u) = e^{-(1-a)u} (1 - e^{-u})^a$. The bank of a CM family is built as in Sect. 3: poles $\theta_k = e^{-u_k}$ with $u_k = u_{10} e^{kh}$, residues $c_k(a) = \sigma_a h u_k m_a(u_k)$, the nodes below the grid lumped onto θ_0 and matched at $r = 1$, and the nodes above it collected in the delay mode.

Theorem 9 (Fixed-pole banks for CM families) *Let $\{k_a\}$ be a CM family and $\hat{k}_a$ the weights realized by its bank.*

- (a) *Propositions 2 and 4, Corollary 3 and Theorems 5 and 6 hold with $k_a(r)$ in place of $g_r(a)$. The parameter a need not be an order. All residues have the sign of σ_a , so parts (i) and (ii) of Proposition 8 apply to every member.*
- (b) *Assume that there are constants $0 < d < \pi/2$, $0 < \beta_- \leq \beta_+$, $\gamma > 0$, $0 < c_- \leq c_+$, C_∞ and $u_* > 0$ such that, for all $a \in \mathcal{A}$:*
 - (H1) *$f_{a,r}(x) = e^x m_a(e^x) e^{-(r-1)e^x}$ is analytic in the strip $|\operatorname{Im} x| < d$ and tends to zero as $|\operatorname{Re} x| \rightarrow \infty$ in it, and*

$$L_d = \sup_{a \in \mathcal{A}, r \geq 1, |y| < d} \frac{\int_{\mathbb{R}} |f_{a,r}(x + iy)| dx}{\int_{\mathbb{R}} f_{a,r}(x) dx} < \infty;$$

- (H2) $c_- u^{\beta_a - 1} \leq m_a(u) \leq c_+ u^{\beta_a - 1}$ on $(0, u_*]$, with $\beta_a \in [\beta_-, \beta_+]$;
(H3) $m_a(u) \leq C_\infty e^{-\gamma u}$ on $[u_*, \infty)$.

Let $\vartheta = \min_{\beta_- \leq \beta \leq \beta_+} \int_0^{u_*} t^{\beta-1} e^{-t} dt$, $C_{\text{lo}} = c_+ / (c_- \beta_- \vartheta)$ and $C_{\text{hi}} = C_\infty / ((1 + \gamma) c_- \vartheta)$. Then, for $u_{\text{lo}} \leq u_*$, $u_{\text{hi}} \geq \max(u_*, 1, \beta_+ \ln 2)$ and $1 \leq r < R$,

$$\left| \frac{\hat{k}_a(r)}{k_a(r)} - 1 \right| \leq \frac{2L_d}{e^{2\pi d/h} - 1} + C_{\text{lo}}((R-1)u_{\text{lo}})^{1+\beta_a} + C_{\text{hi}} e^{-(1+\gamma)u_{\text{hi}}}. \quad (27)$$

- (c) For $0 < \varepsilon \leq \min(1, 4C_{\text{lo}})$ let $h = 2\pi d / \ln(1 + 4L_d/\varepsilon)$, $u_{\text{lo}} = \min(u_*, (\varepsilon/4C_{\text{lo}})^{1/(1+\beta_-)} / (R-1))$, $u_{\text{hi}} = \max(u_*, 1, \beta_+ \ln 2, \ln(4C_{\text{hi}}/\varepsilon)/(1+\gamma))$ and $K = \lceil \ln(u_{\text{hi}}/u_{\text{lo}})/h \rceil + 1$. Then the relative weight error is at most ε for every $a \in \mathcal{A}$ and $1 \leq r < R$, and

$$K = O(\ln(1/\varepsilon) \ln(R/\varepsilon)) \quad (\varepsilon \rightarrow 0, R \rightarrow \infty), \quad (28)$$

uniformly in a .

The proof is given in Appendix F. Part (a) holds because the earlier proofs use only the state equations of the bank and the form $\hat{k}_a(r) = \delta_{r1} c_d(a) + \sum_k c_k(a) \theta_k^{r-1}$. Part (b) combines the error bound of the trapezoidal rule for functions analytic in a strip [52, Thm. 5.1] with the two tail residuals of Sect. 5. For the GL family, (H1)–(H3) hold for every $d < \pi/2$, with $\beta_a = 1 + a$, $\gamma = 1 - \max a$ and $u_* = 1$. The constant L_d grows like $(\cos d)^{-(1+\max a)}$ as $d \rightarrow \pi/2$ (Sect. 7.7). The quadrature term (23) of the rule is the sharper limit $d \rightarrow \pi/2$ for this family: it has π^2/h in the exponent and an algebraic prefactor. With $h \approx \pi^2 / \ln(1/\varepsilon)$, (28) becomes $K \approx \ln(1/\varepsilon) \ln(R/\varepsilon) / \pi^2$ to leading order. A fit over 35 rule designs gives 0.106 for this coefficient, against $1/\pi^2 = 0.101$ (Sect. 7.7).

Example 1 (Distributed and time-varying distributed order) For a probability measure ν on orders of one sign, $k(r) = \int g_r(a) d\nu(a)$ is again a CM kernel, with density $\int \sigma_a m_a d\nu(a)$. Its bank has the same poles and the residues $\int c_k(a) d\nu(a)$. Because all weights have one sign, the relative weight error is at most $\sup_a \varepsilon_R(a)$. If the distribution ν_n changes with time, the output schedule weights the whole memory with ν_n (A-type), and the input schedule weights each sample with the distribution at its entry (B-type). Both still cost $O(K)$ per sample. For distributions of mixed sign the structural statements remain exact, and the error is relative to $\int |g_r(a)| d\nu(a)$.

Example 2 (Tempering) For a fixed $\lambda > 0$, the tempered Grünwald weights $e^{-\lambda r} g_r(a)$ [54] (up to a change of the lag-zero weight) have the density $e^{-\lambda} m_a(u - \lambda)$ on $u > \lambda$. The bank with poles $e^{-\lambda} \theta_k$, residues $e^{-\lambda} c_k(a)$ and delay residue $e^{-\lambda} c_d(a)$ realizes $e^{-\lambda r} \hat{g}_r(a)$ exactly. Its relative weight error, its type and its design rule are therefore those of the untempered bank. A tempering parameter that varies in time moves the poles. By Theorem 6, switching it with a fixed state then cannot give the A- or B-type exactly. A fixed bank for a range of λ is possible, but its density has a branch point at $u = \lambda$, (H1) fails, and the trapezoidal rule converges only algebraically.

Remark 5 (Continuous time) By Bernstein’s theorem, the completely monotone functions are the Laplace transforms of nonnegative measures [53]. For kernels $k_a(t) = \sigma_a \int_0^\infty e^{-\xi t} m_a(\xi) d\xi$ the continuous-time bank of Sect. 3 gives the same structural statements, with e^{Ft} in place of F^n . Examples are the power laws of (10), their distributed-order mixtures, and the Prabhakar kernels in their completely monotone parameter range [55].

7 Numerical results

7.1 Setup

Unless stated otherwise, $T_s = 1$ ms. Discrete-time references are the literal definitions (2)–(5), evaluated in $O(N^2)$ and checked against (7). Continuous-time references are (8) and (9). Errors are relative RMS errors over the post-switch window, or over the whole record.

To separate what switching does from how well each frozen-order member approximates s^α , we also report a *definitional error*: the distance of a switched realization from its *own* A-, B-, D- or E-type, built from the same family of frozen-order systems. For a piecewise-constant order with value set V and masks $M_v = [\alpha = v]$, the own A-type is $y_n = (R_{\alpha_n} u)_n$ and the own B-type is $y = \sum_{v \in V} R_v(u \cdot M_v)$. The own D- and E-types follow from (7).

The baselines are Oustaloup approximations [12] on $[10^{-3}, 10^5]$ rad/s with $N = 3, \dots, 20$ (state count $S = 2N + 1$). They are used in modal (parallel) and first-order-cascade form, discretized by zero-order hold, with the state kept on every coefficient swap. Seeds are fixed, and every table and figure is produced by one script per experiment.

7.2 Which definition does a switched realization track?

Continuous time.

Figure 2 and Table 3 show a unit step with one order switch at $t = 2$ s, at equal state count. The fixed-pole bank with output schedule tracks the A-type and the one with input schedule tracks the B-type, with errors down to 10^{-4} – 10^{-5} . Before the switch, all realizations have similar errors (10^{-3} – 10^{-4}), so the differences come from the switch. The hot-swapped parallel Oustaloup approaches the A-type only slowly. The cascade form tracks neither type: its error stays between 0.16 and 4.6 and does not decrease with S .

One negative result motivates the discrete-time bank. The continuous-time bank with input schedule fails for a smooth *derivative* order (relative error about 12), because modes faster than $1/T_s$ respond to every step of the sampled order.

Discrete time.

The discrete bank (32 modes and the delay mode) was compared with the GL definitions over 7 order profiles and 3 inputs (step, white noise, multisine). The total error of the output schedule against the A-type was 2.5×10^{-6} to 2.0×10^{-3} , and that of the input schedule against the B-type 1.9×10^{-6} to 1.6×10^{-3} . Smooth derivative orders are included, so the continuous-time failure disappears. Table 4 gives the definitional

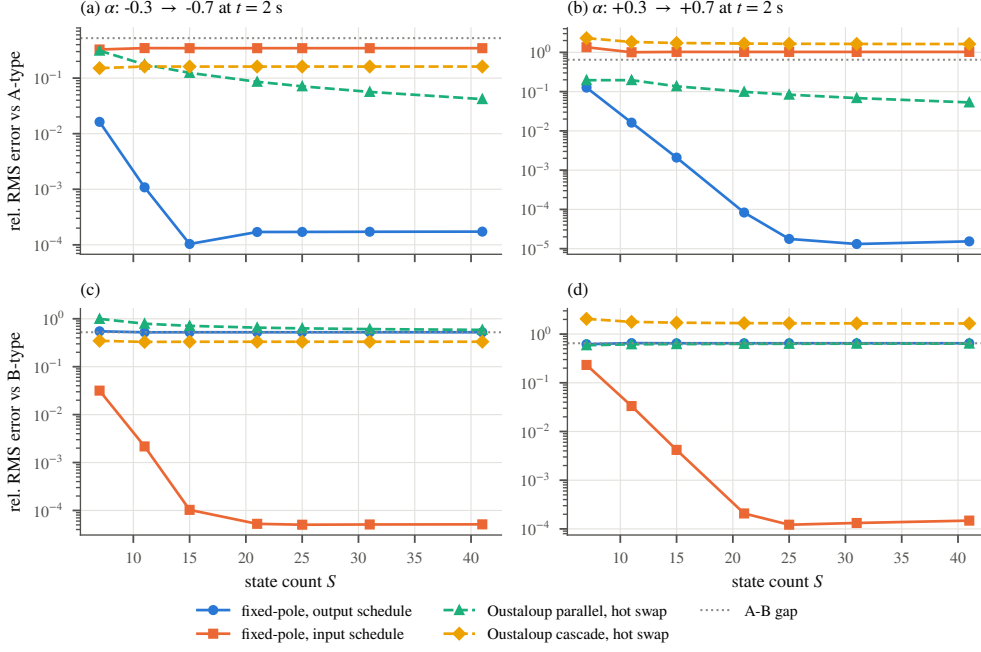


Fig. 2 Continuous time, unit step, one order switch at $t = 2$ s. Relative RMS error after the switch against the A-type (top) and B-type (bottom) closed forms versus state count. The dotted line is the distance between the two definitions. Every series has its own marker and line style

Table 3 Continuous time, step input: relative RMS error after a single switch. FP-out and FP-in are the fixed-pole bank with output and input schedule; Ou-par and Ou-cas are the hot-swapped Oustaloup forms. Arrows show $S = 15 \rightarrow 41$; the cascade column is at $S = 41$

$\alpha_1 \rightarrow \alpha_2$	A-B gap	FP-out vs A	FP-in vs B	Ou-par vs A	Ou-cas, A / B
$-0.3 \rightarrow -0.7$	0.52	$1.0\text{e-}4 \rightarrow 1.7\text{e-}4$	$1.0\text{e-}4 \rightarrow 5.1\text{e-}5$	$0.12 \rightarrow 0.042$	$0.16 / 0.33$
$+0.3 \rightarrow +0.7$	0.65	$2.1\text{e-}3 \rightarrow 1.5\text{e-}5$	$4.2\text{e-}3 \rightarrow 1.5\text{e-}4$	$0.14 \rightarrow 0.053$	$1.6 / 1.7$
$-0.5 \rightarrow +0.5$	0.81	$1.1\text{e-}3 \rightarrow 4.3\text{e-}5$	$5.1\text{e-}4 \rightarrow 3.1\text{e-}5$	$0.23 \rightarrow 0.085$	$4.6 / 1.6$
$+0.5 \rightarrow -0.5$	1.35	$9.8\text{e-}5 \rightarrow 3.4\text{e-}5$	$9.4\text{e-}4 \rightarrow 8.0\text{e-}6$	$0.25 \rightarrow 0.080$	$0.24 / 0.87$

On the fixed band $[10^{-3}, 10^5]$ rad/s used here, the fixed-pole bank reaches its tail floor near $S = 15$. The rule of Sect. 5 chooses the band from the tolerance instead.

errors. For the bank they are zero, or zero to rounding, as Proposition 2 predicts. For the hot-swapped Oustaloup forms they are not, and the cascade form reaches 3×10^3 .

7.3 Can a state map repair a moving-pole realization?

Setup.

For six single switches (at sample 2000, window 2000), we fitted Φ by truncated least squares on a mixed training ensemble of 4 classes with 200 inputs each (white,

Table 4 Discrete time, piecewise-constant orders: definitional error (distance from the realization’s own type). DFP is the discrete fixed-pole bank; the Oustaloup forms have $N = 15$ ($S = 31$)

Realization	vs own A-type	vs own B-type
DFP, output schedule	0 (exact)	equals the A–B gap
DFP, input schedule	equals the A–B gap	$\leq 10^{-14}$
Oustaloup parallel, hot swap	4×10^{-6} to 0.13	7×10^{-3} to 1.0
Oustaloup cascade, hot swap	7×10^{-3} to 3.0×10^3	0.03 to 4.8×10^2

Table 5 State maps for the Oustaloup forms: worst A-type definitional error over 6 switches and 5 input classes (including the step), and the largest norm of the minimum-norm map

S	form	hot swap	LS, float64	minimum-norm map		
				float64 ($\ \Phi\ _2$)	float32	24-bit
7	parallel	0.86	$2.8\text{e-}2$	$5.0\text{e-}2$ (3e4)	$5.0\text{e-}2$	$5.0\text{e-}2$
15	parallel	0.32	$2.1\text{e-}2$	$4.4\text{e-}4$ (5e4)	$4.3\text{e-}4$	$1.0\text{e-}2$
31	parallel	0.14	$4.7\text{e-}7$	$1.7\text{e-}7$ (5e3)	$9.8\text{e-}6$	$3.6\text{e-}3$
7	cascade	$1.1\text{e}4$	$4.4\text{e-}2$	$4.4\text{e-}2$ (9e2)	$4.4\text{e-}2$	2.7e1
15	cascade	$1.1\text{e}4$	$1.0\text{e-}3$	$3.2\text{e-}4$ (8e3)	$8.1\text{e-}1$	$4.3\text{e}2$
31	cascade	$1.1\text{e}4$	$6.2\text{e-}7$	$6.2\text{e-}7$ (4e4)	$2.6\text{e-}1$	$8.1\text{e}2$

The fixed-point test rounds Φ to a single Q format; per-row scaling might do better. The fixed-pole bank needs $\Phi = I$ and has zero definitional error.

piecewise-constant, multisine, Brownian). We tested on fresh ensembles of 150 inputs per class and on a unit step, sweeping the truncation threshold. Besides the most accurate map, we report a robust *minimum-norm* map: the map of smallest norm whose validation error is within twice the best. The maps were then rounded to float32 and to 24-bit fixed point.

Results.

Figure 3 and Table 5 report the worst case over the switches and the input classes. In float64 the error falls from 0.14 (parallel, hot swap) and 1.1×10^4 (cascade, hot swap) to 1.7×10^{-7} and 6.2×10^{-7} at $S = 31$. The map is, however, a dense $S \times S$ matrix, with norm between 10^3 and 5×10^4 even for the minimum-norm choice. Rounded to 24 bits, the map loses 3.5 to 4.3 orders of magnitude of accuracy in the parallel form and 7.4 to 9.1 in the cascade form for $S \geq 21$. In the cascade form the 24-bit error exceeds 1, and even float32 rounding raises it to 0.26–0.81 for $S \geq 15$.

With a smooth derivative order quantized to 41 levels (80 grid crossings, parallel form), a chain of maps reduced the step-input error from 4.1×10^{-2} to 1.2×10^{-4} ($S = 15$). The fixed-pole bank needs no map, and its definitional error is exactly zero. For the $P = 1024$ order levels of Sect. 8, maps between neighbouring levels alone would need about 56 Mbit of storage at $S = 33$ and 25 bits per entry, against 0.78–1.08 Mbit for the whole coefficient table of the bank.

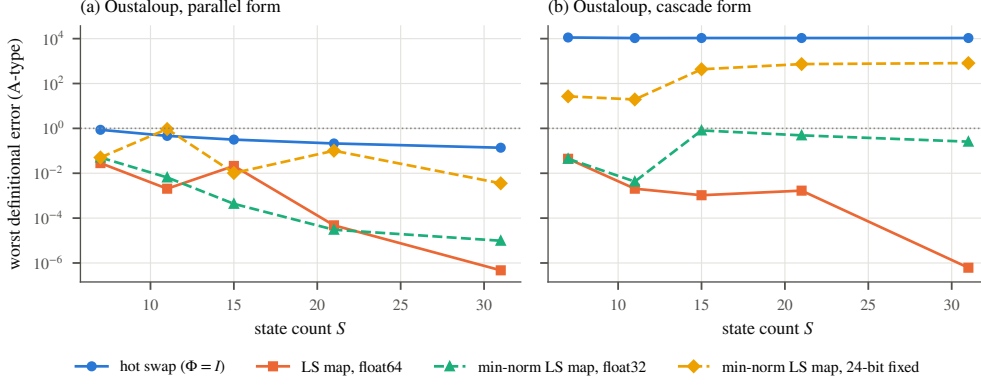


Fig. 3 Worst definitional error (A-type) after one switch, over six switches and five input classes including the step. Hot swap versus least-squares state maps in float64, and the minimum-norm map rounded to float32 and to 24-bit fixed point. The dotted line marks an error of 1

7.4 Design rule on a sealed holdout

Term validation.

Each term of Sect. 5 was first isolated by making the other two negligible (Fig. 4a,b). The ratio of measured to predicted error was 0.49–1.05 in discrete time ($n = 45$) and 0.99–1.01 in continuous time ($n = 66$). The model never underestimates a term by more than 5%.

Holdout.

We drew 400 random specifications (200 discrete, 200 continuous) from a fixed seed:

- $\varepsilon \in [10^{-5}, 10^{-2}]$;
- $R \in [10^2, 10^5]$, or bands of 0.5 to 6 decades;
- random order ranges inside $[-0.95, 0.95]$.

The specifications were written with their SHA-256 digest before any evaluation and then evaluated once. Every specification was met: 200 of 200 in both domains. The one-sided 95% Clopper–Pearson lower bound [56] on the pass rate is 98.5% for each domain. The achieved error was a median 0.41 (discrete) and 0.50 (continuous) of the requested ε , with maxima 0.67 and 0.72. Let $K_{\min}$ be the smallest K on the same grid endpoints that still meets ε , found by bisection. The rule exceeded it by a median of one mode and at most three. Table 6 lists typical sizes: each decade of memory adds about three modes, and each decade of accuracy seven to ten.

7.5 Four types from one bank

Definitions.

The literal recursions (4), (5) agree with $W_A(-\alpha)^{-1}$ and $W_B(-\alpha)^{-1}$ to 1.4×10^{-13} . The dual compositions ($A \circ D$, $D \circ A$, $B \circ E$, $E \circ B$) return the identity to 10^{-12} . The

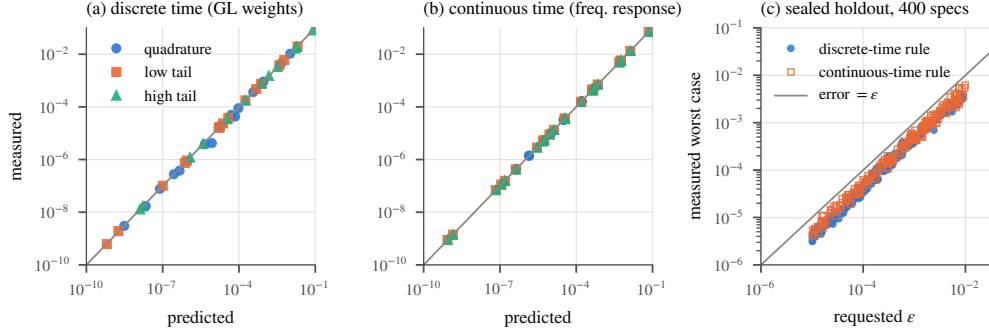


Fig. 4 Analytic error model. (a, b) Each term isolated, measured against predicted. (c) Sealed holdout: measured worst-case error against the requested tolerance for 400 random specifications

Table 6 Bank size K from the rule for $\alpha \in [-0.9, 0.9]$. Discrete time: memory length R in samples (plus the delay mode). Continuous time: band width in decades

ϵ	discrete time, R			continuous time, band		
	10^3	10^4	10^5	2 dec.	4 dec.	6 dec.
10^{-2}	14	17	19	11	14	16
10^{-3}	21	24	27	18	22	26
10^{-4}	29	32	36	27	32	37
10^{-5}	38	42	46	38	44	50

non-dual compositions ($A \circ A$, $B \circ B$, $A \circ B$, $B \circ A$, $A \circ E$, $B \circ D$, $D \circ E$, $D \circ D$) differ from the identity by 10^{-2} to 2.4×10^3 .

Bank.

A bank designed for $\epsilon = 10^{-5}$, $|\alpha| \leq 0.95$ and $R = 4000$ ($K = 41$) was run in all four types over 5 order profiles and 3 inputs (Table 7). The error against the literal definitions is at most 6.9×10^{-5} . The definitional error is at rounding level, and the duality holds in the bank to 2×10^{-14} . The largest output errors occur for derivatives with smooth inputs, where the weights cancel (Corollary 3).

Inverse stability.

Over the 200 discrete-time holdout designs, checked at 25 orders spanning each design range, the inverse was stable at every order in 200/200 designs with the DC floor and in 101/200 without it. Figure 5b shows a D-type integral of order 0.95 driven by white noise for 2×10^5 samples ($\epsilon = 10^{-2}$, $R = 1000$, $K = 15$). Without the floor the output RMS grows to 1.9×10^{21} ; with the floor it stays below 0.36.

Table 7 One bank ($K = 41$) in four types: error against the literal GL definitions (15 cases), and against the bank’s own type (the 12 piecewise-constant cases)

Type	vs literal definition	vs own type (definitional)
A	9.5×10^{-8} to 6.9×10^{-5}	$\leq 2.6 \times 10^{-14}$
B	3.1×10^{-8} to 1.2×10^{-5}	$\leq 1.0 \times 10^{-14}$
D	2.0×10^{-8} to 6.5×10^{-5}	$\leq 2.3 \times 10^{-14}$
E	2.2×10^{-8} to 5.5×10^{-5}	$\leq 2.8 \times 10^{-13}$

Which type does a hot swap implement?

Using the own-type references, we asked which of the four types each hot-swapped realization is closest to, over 8 cases (four piecewise-constant order profiles, step and white-noise inputs) for $N = 7$ and $N = 15$:

- the fixed-pole bank is closest to A (output schedule) or B (input schedule) in all cases, at distances $\leq 2.1 \times 10^{-14}$;
- the parallel Oustaloup form is closest to the A-type in all cases, at distances up to 0.42;
- for the cascade form with $\omega_h = 10^3$ rad/s, the closest type is A, B, D and E in two cases each, so the cascade has no consistent type; with $\omega_h = 10^5$ rad/s it is A or B, at distances up to 1.3×10^3 ;
- with $\omega_h = 10^5$ rad/s, above the Nyquist frequency, the zero-order-hold members of the integral approximations have zeros outside the unit circle (moduli 3 to 41), and their own D- and E-types cannot even be formed in 6 of 8 cases.

Relaxation equations.

For ${}^T\Delta^\alpha y + \lambda y = u$ (Remark 1; $K = 40$, $n = 4000$), the bank matched the triangular GL solutions to 1.7×10^{-5} ($\lambda = 1$) and 4.9×10^{-7} ($\lambda = 100$). The solutions of different types differed by up to 0.40 for $\lambda = 1$ (Fig. 5a), so the choice of type changes the solution visibly. For $\lambda = 100$ the term λy dominates and the types differ by at most 0.02.

7.6 Operator-norm error

Two rule designs with $R = N = 1000$ and $|\alpha| \leq 0.95$ were used: $\varepsilon = 10^{-3}$ ($K = 22$) and $\varepsilon = 10^{-6}$ ($K = 49$). For each, the realized and literal operators of all four types were formed as dense 1000×1000 matrices for eleven order sequences:

- the constants ± 0.5 and the constant order with the largest error row sum;
- three single switches, a random piecewise-constant sequence, an independent uniform order at every sample, and a smooth zero crossing;
- two adversarial sequences that take, at every lag, the order with the largest error on a grid of 380 orders.

The matrices reproduce the streaming bank to rounding. Table 8 compares the measured norms with Theorem 5.

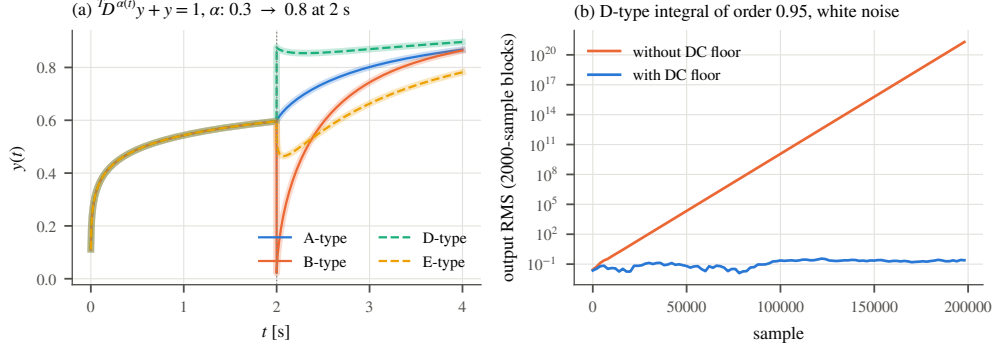


Fig. 5 (a) Relaxation equation of all four types with one bank; thin lines are the bank, wide light lines the triangular GL solutions. (b) D-type integral of order 0.95 (inverse of the 0.95 derivative bank) on white noise, with and without the DC floor

Table 8 Theorem 5 on dense operators ($N = 1000$, eleven order sequences). Ratios of measured to bounded quantities; ε_R is taken over the orders of each sequence, except in the last row, where it is the design value

Types	Quantity	$\varepsilon = 10^{-3}$, $K = 22$	$\varepsilon = 10^{-6}$, $K = 49$
A, B	$\ E\ _p / \text{bound (a)}, p = 1, \infty$	0.28–1.00	0.45–1.00
A, B	$\ E\ _2 / \text{bound (a)}$	0.13–0.87	0.16–0.89
A, B	$(\ E\ _p / \ W\ _p) / \varepsilon_R, p = 1, \infty$	0.04–0.39	0.03–0.45
D, E, integrals	$\ E\ _\infty / \text{bound (c)}$	0.23–0.31	0.006–0.23
D, E, all	$\max(\ E\ _\infty / \ W\ _\infty) / \varepsilon_R$	22	17

For the A- and B-types no norm exceeded its bound, and the bounds of part (a) were attained to four digits by the constant and the adversarial sequences. The relative operator error in ℓ^1 and ℓ^∞ was at most $0.45 \varepsilon_R$. The error kernel is much smaller than ε_R times the weights: at $\varepsilon = 10^{-3}$, $\rho_{\text{row}} = 0.031$, against $\varepsilon_R \sup_a \sum_r |g_r(a)| = 0.27$. The weight error is relatively largest where the weights are small.

For the recursive integrals (the two sequences with only negative orders), the measured ℓ^∞ error was 0.23–0.31 of the bound of part (c) at $\varepsilon = 10^{-3}$, and the certificate computed from one step response of the bank gave the same values. The amplification predicted by Remark 2 is real: the relative error of the D- and E-types reached 22 times the relative weight error of the design. For recursive derivatives it stayed below ε_R , and the bound was conservative by factors of 14 to 7×10^4 , which is the condition number κ_∞ (71 to 1445 here) times the margin of part (b).

7.7 Completely monotone families

Strip constant and quadrature bound.

For the GL family, L_d was evaluated over 13 orders in $[-0.95, 0.95]$ and lags $r = 1$ to 10^5 . It is 2.0, 7.2, 31.7 and 175 for $d = 0.8, 1.2, 1.4$ and 1.5 , which follows $(\cos d)^{-1.95}$.

Minimized over d , the quadrature term of (27) exceeded the measured worst error of the infinite trapezoidal rule (13 orders, 10 lags, 8 grid offsets) by factors of 1.9 to 3.7 for $0.3 \leq h \leq 1$. The sharper GL term (23) agreed with the measurement to within 1%. The general theorem thus loses less than a factor of four against the analysis specific to GL, which costs two to three modes at $K \approx 40$.

Bank size.

Over 35 rule designs with $R = 10^2, \dots, 10^6$ and $\varepsilon = 10^{-2}, \dots, 10^{-8}$ (K from 12 to 90), the least-squares fit

$$K \approx 0.57 + 1.06 \ln(1/\varepsilon) + 0.53 \ln R + 0.106 \ln(1/\varepsilon) \ln(R/\varepsilon)$$

reproduces every K to within 0.7 modes. The coefficient of the product term is within 5% of the value $1/\pi^2$ predicted by Theorem 9.

Distributed and tempered kernels.

For three time-varying distributed orders (Example 1; uniform densities of width 0.3 whose centre moves smoothly in $[0.15, 0.75]$ or in $[-0.75, -0.15]$, or switches from -0.5 to 0.5), one bank designed for $\varepsilon = 10^{-5}$ ($K = 39$) was run with the averaged residues. Its kernel error was below the largest error of its members in all three cases (at most 2.6×10^{-6}). The relative operator error of the A- and B-types was at most 1.8×10^{-7} , and the streaming bank matched the dense operators to 3×10^{-16} . For the tempered weights with $\lambda = 0.01$ (Example 2), the relative weight error equals that of the untempered bank, 3.877×10^{-6} , to rounding.

8 Fixed-point realization

The design point is $R = 4000$, $\varepsilon = 10^{-4}$ and $|\alpha| \leq 0.95$, giving $K = 32$ plus the delay mode. Orders come from a grid of $P = 1024$ levels, $\alpha_i = 0.95(2i/(P-1) - 1)$, so $\Delta\alpha = 1.9 \times 10^{-3}$.

8.1 Arithmetic in sample units

The core runs in sample units ($T_s = 1$), where $\hat{g}_0 = 1$ for every order. The inversion (21) then needs no divider: $v_n = x_n - \eta_n$. Physical units follow from one gain $T_s^{-\alpha_n}$, applied at the output for the A- and E-types and at the input for the B- and D-types. This follows directly from (2)–(5); for the E-type, substitute $z_n = T_s^{-\alpha_n} \zeta_n$.

The number formats are as follows:

- *Signals*: W_S -bit two's complement with F fractional bits.
- *States*: $W_{ST} = W_S + 13$ bits. Each mode stores its state scaled by a power of two 2^{G_k} (one gain per schedule) so that its own bound uses the full word. Without this scaling, the input-scheduled slow modes lose their increments $c_k v_n$, which lie far below the signal LSB; the B- and E-type errors were then 2×10^{-2} to 0.66.
- *Coefficients*: a W_M -bit mantissa with a 7-bit shift.

Algorithm 1 One sample of the bank, all four types. In hardware the coefficient row is a ROM word addressed by the order index (mirrored for D and E), so an order or type change costs nothing

Require: type $T \in \{A, B, D, E\}$, order α_n , input x_n ; states $s_0, \dots, s_{K-1}, s_d$

```

1:  $a \leftarrow \alpha_n$  if  $T \in \{A, B\}$ , else  $a \leftarrow -\alpha_n$ ; read  $(c_d, c_0, \dots, c_{K-1})$  for order  $a$ 
2: if  $T \in \{A, D\}$  then ▷ output schedule
3:    $\eta \leftarrow c_d s_d + \sum_k c_k s_k$ 
4: else ▷ input schedule
5:    $\eta \leftarrow s_d + \sum_k s_k$ 
6: end if
7: if  $T \in \{A, B\}$  then  $v \leftarrow x_n$ ;  $y_n \leftarrow v + \eta$ 
8: else  $v \leftarrow x_n - \eta$ ;  $y_n \leftarrow v$  ▷ no divider in sample units
9: end if
10: if  $T \in \{A, D\}$  then  $s_k \leftarrow \theta_k s_k + v$ ;  $s_d \leftarrow v$ 
11: else  $s_k \leftarrow \theta_k s_k + c_k v$ ;  $s_d \leftarrow c_d v$ 
12: end if
```

Products are rounded as $\text{mulsh}(a, M, S) = (aM + 2^{S-1}) \gg S$. Algorithm 1 gives one sample of the core in exact arithmetic; the bit-accurate golden model and the RTL use the same sequence with the formats above.

8.2 Dynamic range: two operating envelopes

For $|x| \leq 1$ over 4000 samples, the frozen-order ℓ_1 bound on the output is 2696. For the A-type this bound holds under any switching, because each output uses the weights of a single order; for the B-type the switching stress tests stayed within it. For the D- and E-types, switching from an integral to a derivative order drives the output to 2.2×10^6 under a DC input, 820 times the frozen-order bound. The new inverse acts on a history that has grown under the old order. This is the behaviour of the recursive definitions themselves, not an artefact of the bank. We therefore size two envelopes with a factor-two margin: 14 integer bits for the forward types only (A/B), and 24 integer bits for all four types (A/B/D/E).

8.3 Quantization and inverse stability

Quantization can undo the DC floor. Coefficients and pole steps are dyadic rationals, so the signs of $\hat{H}_q(\pm 1)$ of every quantized ROM row can be evaluated exactly in rational arithmetic, which we did. With the floating-point floor alone:

- in the A/B configuration, 64 of the 512 positive-order rows have $\hat{H}_q(1) \leq 0$, the worst having its dominant inverse pole at $1 + 4.8 \times 10^{-6}$ (e-folding time 2.1×10^5 samples);
- in the A/B/D/E configuration, 51 of 512 rows have $\hat{H}_q(1) \leq 0$, the worst ($\alpha = 0.855$) having its pole at $1 + 1.9 \times 10^{-9}$ (e-folding time 5.2×10^8 samples, about 9 minutes at 1 MS/s).

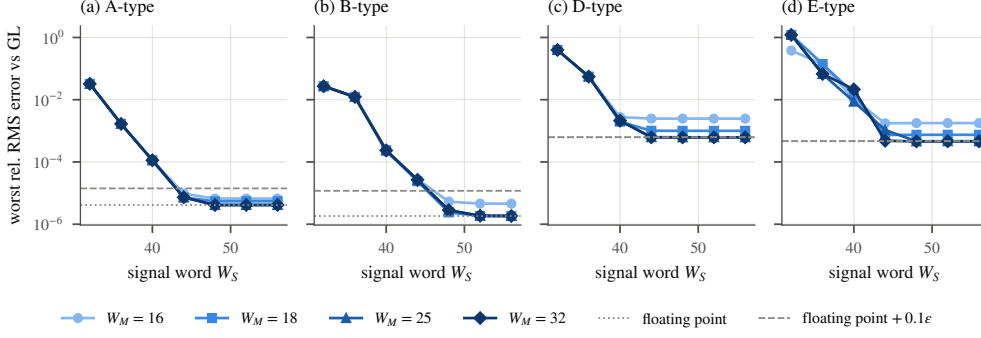


Fig. 6 Word length, A/B/D/E envelope: worst relative RMS error against the literal definitions versus signal word W_S (state word $W_S + 13$), for four mantissa widths. Dotted: floating-point bank; dashed: acceptance limit

$\hat{H}_q(-1)$ was positive in every row. The *quantization-aware* DC floor raises the delay mantissa in LSB steps until $\hat{H}_q(1)$ is at least the floating-point value $\hat{H}(1)$. It brings both counts to zero without any measurable change in accuracy.

8.4 Word length

We swept $W_S \in \{32, \dots, 56\}$ and $W_M \in \{16, 18, 25, 32\}$ (Fig. 6). The test set comprised 3 order profiles (random piecewise, a new order every sample, a sign-changing switch), 3 inputs, and all types of the envelope. A configuration was accepted if, for every type, its error against the literal definition was at most the floating-point bank's error plus 0.1ε . The smallest accepted configurations were:

- A/B: $W_S/W_{ST}/W_M = 36/49/16$;
- A/B/D/E: $48/61/25$.

With an 18-bit mantissa the D- and E-types stall near 10^{-3} , because the inversion is ill-conditioned: the low-frequency gain of a derivative bank is about $R^{-\alpha}$. The floating-point bank itself has D- and E-type output errors up to 6×10^{-4} at this design point, although its weight error is 10^{-4} . This is the cancellation effect of Corollary 3, amplified by the inversion.

8.5 RTL, parity and resources

The core (`vo_bank_core.v`) is a non-pipelined, time-multiplexed state machine with synchronous coefficient, pole and gain ROMs. It takes $4K + 8 = 136$ clock cycles per sample. Order index and type are inputs with every sample. An order change only changes a ROM address: there is no reload and no extra cycle.

The test vectors cover constant, piecewise and per-sample orders, sign-changing switches, full-scale inputs, and type changes at run time. The type changes exercise the datapath; a type change is not itself a VO operation. Icarus Verilog simulation matched the golden model bit for bit on all 94 208 vectors (31 744 for A/B, 62 464 for

Table 9 RTL configurations: bit-exact parity with the golden model and Yosys `synth_xilinx` (xc7) resource estimates. Vivado results and $f_{\max}$ are pending

Envelope	$W_S/W_{ST}/W_M$	vectors (mismatches)	LUT	FF	DSP48E1	RAMB36 eq.
A/B	36/49/16	31 744 (0)	4 171	1 882	11	22
A/B/D/E	48/61/25	62 464 (0)	6 009	2 360	29	29.5

A/B/D/E). Verilator lint reported only intentional width extensions and truncations. Table 9 lists Yosys estimates for a 7-series FPGA.

[Board measurements (to be added): bitstream-level parity on the same vectors, Vivado utilization and $f_{\max}$, sample rate $f_{\max}/136$, per-sample order changes, and a long run ($\geq 10^7$ samples) of the D-type at the worst quantized row. These will be measured with the new core; no measurement from [23] is reused.]

9 Application: streaming multifractional Brownian motion

Multifractional Brownian motion lets the Hurst exponent vary in time [57], and its definitions are in general not equivalent [58]. The Riemann–Liouville form (RL-mBm) [59, 60] is a VO integral of white noise of order $H(t) + \frac{1}{2}$, with the current order applied to the whole past. It is thus an A-type integral. Applying the order at the time each increment enters gives the B-type variant. Up to the discretization of the kernel and a per-increment factor $\sqrt{2H_j} \Gamma(H_j + \frac{1}{2})$, this is the memory-multifractional Brownian motion (MMFBM) of Wang et al. [26], $X(t) = \int_0^t \sqrt{\alpha(s)} (t-s)^{(\alpha(s)-1)/2} dB(s)$ with $\alpha = 2H$. The input-scheduled bank applies that factor as a gain on each input sample. The incremental mBm of Ślęzak and Metzler [61] is defined through the covariance of its increments and shares the property that a change of H acts only on new increments. Other alternatives to mBm include:

- the multifractional processes of Surgailis, built by nonhomogeneous fractional integration of white noise [46], whose covariance is not a local function of $H(t)$ and $H(t')$ as it is for mBm [62];
- the integrated fractional white noise of Sly, proposed to avoid the extreme changes in magnitude of mBm [63].

Conversely, the variable-Hurst fBm of Ryvkina uses the current Hurst value in its kernel and has discontinuous paths wherever H jumps [64], like the A-type. VO operators have been used to synthesize multifractional Gaussian noise before [65], and simulation methods for mBm include [66]. Exact methods for stationary Gaussian processes such as circulant embedding [67] rely on stationarity and do not apply directly when H varies.

In sample units, the discrete processes are

$$X_n^A = \sum_r g_r(-a_n) \xi_{n-r}, \quad X_n^B = \sum_j g_{n-j}(-a_j) \xi_j, \quad a_n = H_n + \frac{1}{2},$$

with i.i.d. standard normal ξ . For $H > \frac{1}{2}$ the order exceeds 1, outside the range of the bank. An integer integration can be split off exactly, but only in the right place:

Proposition 10 (Integer split) *Let Σ denote the cumulative sum. Then*

$${}^A\Delta^{-a}\xi = {}^A\Delta^{-(a-1)}(\Sigma\xi), \quad {}^B\Delta^{-a}\xi = \Sigma({}^B\Delta^{-(a-1)}\xi).$$

Proof $(1 - z^{-1})^{-a} = (1 - z^{-1})^{-(a-1)}(1 - z^{-1})^{-1}$ gives $w_m(-a) = \sum_{i=0}^m w_i(-(a-1))$ for every fixed a . For the A-type, apply this at the fixed order a_n of output sample n and exchange the sums. For the B-type, sample j keeps its order a_j :

$$\sum_{j \leq n} \sum_{i \leq n-j} w_i(-(a_j-1))\xi_j = \sum_{m \leq n} \sum_{j \leq m} w_{m-j}(-(a_j-1))\xi_j.$$

The right-hand side is the cumulative sum of ${}^B\Delta^{-(a-1)}\xi$. $\square$

The reversed orders are not exact: the relative errors are 0.1 to 1.3 in our tests. With the split, one bank of order $\frac{1}{2} - H$, which satisfies $|\frac{1}{2} - H| < \frac{1}{2}$, covers all of $0 < H < 1$. It crosses $H = \frac{1}{2}$ seamlessly because the order 0 is exact in the bank.

Results.

Table 10 and Fig. 7 summarize the results. The bank was designed by the rule for $\varepsilon = 10^{-4}$ and $0.05 \leq H \leq 0.95$, giving $K = 22, 24$ and 26 for $n = 1024, 4096$ and 16384 .

Second-order statistics were computed exactly, without Monte Carlo, from the $n \times n$ weight matrix of the generator. The variance agreed with the exact formula to 3.1×10^{-5} and the covariance to 1.7×10^{-6} .

The local Hurst exponent was estimated by generalized quadratic variations [68, 69]; see also [62]. For constant H the estimator itself has a bias of $+0.06$ at $H = 0.2$ and a standard deviation of 0.06 (window 1024). For varying H its mean followed $H(t)$ with RMSE 0.014 to 0.036, equally for both types.

The two types differ sharply at a jump of H from 0.3 to 0.8. The A-type variance jumps from 119 to 1.5×10^5 , and the path is discontinuous: the squared increment at the jump is 1.3×10^5 times the typical one. The B-type is continuous (ratio 1.0), in line with the behaviour reported for MMFBM and incremental mBm [26, 61]. Which definition to use is therefore a modelling decision with visible consequences, as also argued in [26]. The bank makes either choice available at the same cost.

Hardware.

The cores of Sect. 8 generate the same processes bit-exactly from integer noise. For the A-type, the accumulator sits in front of the core and is exact (an integer sum), so the small A/B core suffices. For the B-type, the accumulator follows the core and integrates its low-frequency rounding errors. The B-type therefore benefits from the wider configuration (Table 10, last two rows). Input scales were 2^{-9} for A and 0.25 for B, chosen to keep the accumulated A-type input within the signal range.

Table 10 Streaming mBm with the fixed-pole bank

Quantity	Result
integer split, correct order (A first, B last)	$\leq 9 \times 10^{-15}$
integer split, reversed order	0.1 to 1.3
path vs literal GL, same noise	$A \leq 1.9 \times 10^{-6}$, $B \leq 2.4 \times 10^{-6}$
exact variance / covariance ($n = 2048$)	$\leq 3.1 \times 10^{-5}$ / $\leq 1.7 \times 10^{-6}$ (Frobenius)
local Hurst tracking, varying H (400 paths)	RMSE of mean 0.014 to 0.036, A and B alike
variance before/after a jump of H from 0.3 to 0.8	A $119 \rightarrow 1.5 \times 10^5$; B $119 \rightarrow 119$
cost per sample	bank 47 to 55 MACs (n -independent); direct GL $n/2$ on average
fixed-point core A/B (36/49/16), vs float	A 1.8×10^{-5} , B 8.9×10^{-4}
fixed-point core A/B/D/E (48/61/25), vs float	A 3.8×10^{-6} , B 1.2×10^{-4}

10 Discussion and limitations

Scope of the guarantees.

Proposition 2 and Corollary 3 hold for every order sequence. Proposition 8 is a statement about each *frozen* order. The recursive types under switching form a switched linear system $F - b c(\alpha_n)^\top / \hat{g}_0$, whose stability under arbitrary switching is not implied by the stability of each member. In all our tests the switched D- and E-types stayed bounded, within the stress envelope of Sect. 8, but we have no proof for arbitrary switching.

Output versus weight error.

Theorem 5 turns the weight guarantee into an operator guarantee: for the A- and B-types, the worst-case relative output error over all inputs, in ℓ^1 and ℓ^∞ , is at most ε_R . Particular inputs can still show larger relative errors, because a derivative of a smooth input is much smaller than the input. The relative output error then exceeded ε_R by up to a factor of seven for the bank of Sect. 7.5, and by up to six for the D- and E-types at the design point of Sect. 8. For the recursive integrals the operator error itself is amplified, by up to $\kappa_\infty \approx 2N^a/\Gamma(1+a)$ (Remark 2); a guaranteed design for them must use $\varepsilon/\kappa_\infty$, which adds about $K \ln \kappa_\infty / \ln(1/\varepsilon)$ modes.

Order range.

The bank covers $-1 < \alpha < 1$. Larger orders follow from integer splits such as Proposition 10, which must be placed according to the type.

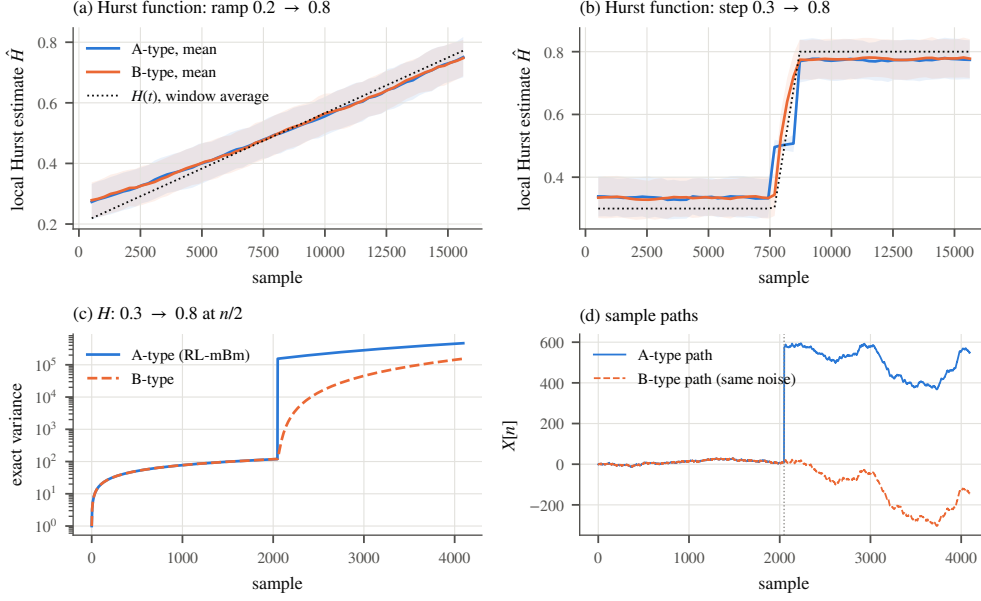


Fig. 7 Streaming mBm. (a, b) Local Hurst estimates (mean $\pm$ one standard deviation over 400 paths) for a ramp and a step of H . (c) Exact variance of the A-type (RL-mBm) and B-type processes for a jump of H at $n/2$. (d) Sample paths driven by the same noise

Fixed point.

We used a single Q format per signal. Block floating point could reduce the D/E word lengths, which are dominated by the switching envelope.

Hardware.

The core is not pipelined. A pipelined version could approach K plus a few cycles per sample. Resources are synthesis estimates, and the board measurements are pending.

Literature search.

The review of Sect. 1.1 used Crossref and arXiv. [TODO: repeat the documented queries in Scopus and Web of Science before submission.]

11 Conclusion

For a finite-dimensional realization of a variable-order operator, the definition it implements is a property of its structure. If the poles move with the order, no state map can make a switched realization exact for the A- or B-type. If the poles are fixed and the order sits in the output or input weights, the realization is the A- or B-type for every order sequence, up to the frozen-order quadrature error. Inverting the same bank gives the D- and E-types with exact duality. The Beta-integral representation makes this exact in discrete time. The weight error bounds the realized operators as a whole: for

the forward types the relative operator error never exceeds it, and for the recursive integrals it is amplified by a computable condition number. Nothing in the construction is specific to Grünwald–Letnikov weights: every kernel family that is completely monotone in the lag, such as distributed orders with a time-varying density, is realized in the same way, with a common pole set whose size grows like $\ln(1/\varepsilon) \ln(R/\varepsilon)$. An analytic rule sizes the bank, and a DC floor, tested by an exact sign, keeps every inverse stable, also after quantization. In hardware the order and the type become a ROM address that can change at every sample. Streaming multifractional Brownian motion shows that the choice between the A- and B-type is a modelling decision with visible consequences, which the bank makes explicit.

Acknowledgements. [TODO.]

Declarations

- **Funding:** [TODO.]
- **Competing interests:** The authors declare no competing interests. [TODO: confirm.]
- **Data availability:** All results reported in this paper are produced by the scripts in the accompanying repository, with fixed seeds; the result files are included.
- **Code availability:** [TODO: public repository URL or archive DOI.] The repository contains the Python library, the experiments, the Verilog core, the testbench and the board test bundle.
- **Author contributions:** [TODO.]
- **Relation to other work:** A constant-order FPGA approximation by the same authors is under review at AEÜ – International Journal of Electronics and Communications [23]. The present paper uses different structures (order-independent poles), a different design rule (analytic, fit-free) and a new core. It reuses no figure, table, text or measurement from that work.

Appendix A Proof of Theorem 6

Write $\mathcal{C}_i = [b_i, F_i b_i, \dots, F_i^{S-1} b_i]$ and $\mathcal{O}_i = [c_i, F_i^\top c_i, \dots, (F_i^\top)^{S-1} c_i]^\top$, both invertible by minimality [70].

(a) For $n \geq m$ the switched output is $d_2 v_n + c_2^\top F_2^{n-m} \Phi x_1[m] + c_2^\top \sum_{m \leq j < n} F_2^{n-1-j} b_2 v_j$. The response of Σ_2 to the whole input differs from it only by $\Phi x_1[m]$ being replaced by $x_2[m] = \sum_{j < m} F_2^{m-1-j} b_2 v_j$. Equality for all inputs and all $n \geq m$ means $c_2^\top F_2^k (\Phi x_1[m] - x_2[m]) = 0$ for all $k \geq 0$. By observability this holds if and only if $\Phi x_1[m] = x_2[m]$ for every input history, that is, if and only if $\Phi F_1^k b_1 = F_2^k b_2$ for $k = 0, \dots, m-1$. Since $m \geq S+1$, this gives $\Phi \mathcal{C}_1 = \mathcal{C}_2$, so Φ is invertible. It also gives $(\Phi F_1 - F_2 \Phi) F_1^k b_1 = \Phi F_1^{k+1} b_1 - F_2^{k+1} b_2 = 0$ for $k = 0, \dots, S-1$, hence $(\Phi F_1 - F_2 \Phi) \mathcal{C}_1 = 0$ and $F_2 = \Phi F_1 \Phi^{-1}$. Conversely, if $F_2 = \Phi F_1 \Phi^{-1}$ and $b_2 = \Phi b_1$, then $\Phi F_1^k b_1 = F_2^k b_2$ for all k .

(b) The B-type behaviour requires that the stored state continue as in Σ_1 : $c_2^\top F_2^k \Phi x = c_1^\top F_1^k x$ for all $k \geq 0$ and every reachable x . By controllability every x is reachable within $S \leq m-1$ samples, so $\mathcal{O}_2 \Phi = \mathcal{O}_1$. Then Φ is invertible, $c_2^\top = c_1^\top \Phi^{-1}$,

and $\mathcal{O}_2(F_2\Phi - \Phi F_1) = 0$ follows in the same way, so $F_2 = \Phi F_1 \Phi^{-1}$. The converse is immediate. $\square$

Appendix B Proof of Proposition 8

N has degree $K + 1$. If $c_d = 0$, then $z = 0$ is a root of N ; it cancels the delay pole and lies inside the disc. We give the argument for $c_d \neq 0$; the case $c_d = 0$ is the same with the pole at 0 removed.

(i) All residues are positive, so $\hat{H}$ decreases from $+\infty$ to $-\infty$ between consecutive poles of $\{0, \theta_0, \dots, \theta_{K-1}\}$. This gives K roots in $(0, \max_k \theta_k)$. On $(\max_k \theta_k, \infty)$, $\hat{H}$ decreases from $+\infty$ to $g_0 > 0$, so there is no root there. On $(-\infty, 0)$, $\hat{H}$ decreases from $g_0 > 0$ to $-\infty$, so there is exactly one root z_* . It lies in $(-1, 0)$ if and only if $\hat{H}(-1) > 0$.

(ii) Between consecutive θ_k the negative residues make $\hat{H}$ rise from $-\infty$ to $+\infty$, which gives $K - 1$ roots. For $z > \max_k \theta_k$,

$$z\hat{H}(z) = g_0 z + c_d + \sum_k c_k (1 + \theta_k/(z - \theta_k))$$

is strictly increasing from $-\infty$ to $+\infty$. Hence there is exactly one root there, and it is smaller than 1 if and only if $\hat{H}(1) > 0$. If $c_d < 0$, then $z\hat{H}(z)$ rises from $c_d < 0$ at $z = 0$ to $+\infty$ at $\min_k \theta_k^-$, giving the last root in $(0, \min_k \theta_k)$. If $c_d > 0$, then $z\hat{H}(z)$ rises from $-\infty$ at $z \rightarrow -\infty$ to $c_d > 0$ at $z = 0$. Counting shows there is exactly one root $z_* < 0$, with $z\hat{H}(z) < 0$ for $z < z_*$. Hence $z_* > -1$ if and only if $-\hat{H}(-1) < 0$, that is, $\hat{H}(-1) > 0$. In all cases the $K + 1$ roots are real, so no complex root can lie outside the disc.

(iii) Integral bank, order $-a$ with $0 < a < 1$. Since $\hat{g}_1 = c_d + \sum_k c_k$ and $\hat{g}_2 = \sum_k c_k \theta_k$,

$$\hat{H}(-1) = g_0 - \hat{g}_1 + \sum_k \frac{c_k \theta_k}{1 + \theta_k} \geq g_0 - \hat{g}_1 + \frac{1}{2} \hat{g}_2.$$

With $g_1 = ag_0$, $g_2 = \frac{1}{2}a(1+a)g_0$ and relative errors at most $\delta = \frac{1}{3}$, the right-hand side is at least $g_0(1-a)(6-a)/6 > 0$. Derivative bank, order a : here $\sum_k c_k \theta_k/(1 + \theta_k) \geq \hat{g}_2$ because $c_k < 0$. Hence $\hat{H}(-1) \geq g_0 - \hat{g}_1 + \hat{g}_2$. With $g_1 = -ag_0$ and $g_2 = -\frac{1}{2}a(1-a)g_0$, this is at least $g_0(1 + \frac{2}{3}a^2) > 0$. $\square$

Appendix C B-type step response

The B-type operator in continuous time processes each input sample with the kernel of its entry order: $y(t) = \int_0^t K(t - \tau, \alpha(\tau))u(\tau) d\tau$ with $K(s, a) = \partial_s F(s, a) = s^{-a-1}/\Gamma(-a)$, taken in the finite-part sense for $a > 0$. For $u = 1$, let $G(\tau) = F(t - \tau, \alpha(\tau))$. Then

$$\frac{dG}{d\tau} = -\partial_s F(t - \tau, \alpha(\tau)) + \partial_a F(t - \tau, \alpha(\tau)) \dot{\alpha}(\tau),$$

so $y(t) = G(0) - G(t) + \int_0^t \partial_a F \dot{\alpha} d\tau$. The finite part removes $G(t) = F(0^+, \alpha(t))$, and $G(0) = F(t, \alpha(0))$. This gives (9). For a piecewise-constant order, $\dot{\alpha}$ is a sum of Dirac impulses and the integral becomes the stated finite sum. In the experiments the smooth case was evaluated by adaptive quadrature after the substitution $s = v^m$ with $m(1 - \alpha_{\max}) \geq 2$, which removes the weak singularity at $s = 0$.

Appendix D Large-lag alias term

For large r the integrand of (14) is concentrated at $u = O(1/r)$, where $(1 - e^{-u})^a \approx u^a$. Hence $f(x) \approx e^{(1+a)x} e^{-\rho e^x}$ with $\rho = r - a$. Its Fourier transform is $\hat{f}(\omega) = \Gamma(1 + a - i\omega)\rho^{-(1+a-i\omega)}$, and $\hat{f}(0) = \Gamma(1 + a)\rho^{-(1+a)}$ is the integral itself. The relative error of the infinite trapezoidal rule is $\sum_{k \neq 0} \hat{f}(2\pi k/h)/\hat{f}(0)$, dominated by $k = \pm 1$. Stirling's formula, $|\Gamma(1 + a - i\omega)| \sim \sqrt{2\pi} \omega^{a+1/2} e^{-\pi\omega/2}$, at $\omega = 2\pi/h$ gives (23). The result does not depend on r , which justifies the name large-lag term.

Appendix E Proof of Theorem 5

(a) Row n of E_A is $(e_n(\alpha_n), \dots, e_1(\alpha_n), 0, \dots, 0)$. It contains a single order, so its absolute sum is at most ρ_{row} . Column j of E_A contains $e_{n-j}(\alpha_n)$ for $n > j$, one entry per lag, each at most $\sup_a |e_{n-j}(a)|$, so its absolute sum is at most ρ_{col} . In E_B column j contains the single order α_j , and row n contains one entry per lag. This gives the four bounds. The ℓ^2 bound follows from $\|E\|_2^2 \leq \|E\|_1 \|E\|_\infty$. For a finite $\mathcal{S}$, let a^* maximize $\sum_r |e_r(a)|$ and a_r maximize $|e_r(a)|$. The constant sequence a^* attains ρ_{row} in the last row of E_A and in the first column of E_B . The sequence $\alpha_n = a_n$ attains ρ_{col} in the first column of E_A , and the reversed sequence $\alpha_n = a_{N-1-n}$ attains it in the last row of E_B .

(b) The off-diagonal entries satisfy $|(E_T)_{n,j}| \leq \varepsilon_R |(W_T)_{n,j}|$ because $n - j < N \leq R$, and the diagonal of E_T vanishes. For any x , $|E_T x| \leq |E_T| |x| \leq \varepsilon_R |W_T| |x|$ entrywise, so $\|E_T x\|_p \leq \varepsilon_R \|W_T\|_p \|x\|_p$. For $p \in \{1, \infty\}$ the induced norm depends only on the absolute values of the entries.

(c) Since $\hat{W}_T = \hat{W}_F(-\alpha)^{-1}$ and $W_T = W_F(-\alpha)^{-1}$,

$$\hat{W}_T - W_T = \hat{W}_T (W_F(-\alpha) - \hat{W}_F(-\alpha)) W_T = -\hat{W}_T E_F^- W_T.$$

Hence $\|E_T\|_p \leq \|\hat{W}_T\|_p \|E_F^-\|_p \|W_T\|_p \leq (\|W_T\|_p + \|E_T\|_p) q$, which rearranges to the first bound. Writing $\|W_T\|_p \leq \|\hat{W}_T\|_p + \|E_T\|_p$ instead gives the second. If all orders are negative, $\hat{W}_F(-\alpha)$ is a derivative bank at every sample: its diagonal $T_s^{\alpha_n}$ is positive, and its off-diagonal entries $\hat{g}_r(-\alpha_n)$, $r \geq 1$, are nonpositive, for $r \geq 2$ because all $c_k < 0$ and for $r = 1$ by assumption. (The DC floor changes $\hat{g}_1$ only by about the DC quadrature error; in the 200 discrete-time holdout designs, $\hat{g}_1 < 0$ at all 4294 checked derivative orders.) Write $\hat{W}_F(-\alpha) = \Delta - L$ with Δ diagonal positive and $L \geq 0$ strictly lower triangular. Then $\hat{W}_T = \sum_{k=0}^{N-1} (\Delta^{-1} L)^k \Delta^{-1} \geq 0$, and for a nonnegative matrix $\|M\|_\infty = \|M \mathbf{1}\|_\infty$. $\square$

Appendix F Proof of Theorem 9

Part (a) is stated after the theorem. For (b), fix a and r , and write $f = f_{a,r}$ and $I = \int f dx = k_a(r)/\sigma_a$. The realized weight is σ_a times the infinite trapezoidal sum $T_h = h \sum_{j \in \mathbb{Z}} f(x_j)$, $x_j = \ln u_{l_0} + jh$, minus two residuals. Let $u_j = e^{x_j}$.

Quadrature. By (H1) and [52, Thm. 5.1], $|T_h - I| \leq 2M/(e^{2\pi d/h} - 1)$ with $M = \sup_{|y| < d} \int |f(x + iy)| dx \leq L_a I$.

Lower bound on I . For $r \geq 2$, (H2) gives $I \geq c_- \int_0^{u^*} e^{-(r-1)u} u^{\beta_a-1} du \geq c_- \vartheta (r-1)^{-\beta_a}$.

Lower tail. The nodes $j < 0$ are replaced by one node at $u_0 = u_{l_0}$ with the same total weight. At lag r the residual is $\sum_{j < 0} h u_j m_a(u_j) (e^{-(r-1)u_j} - e^{-(r-1)u_0})$, which lies between 0 and $(r-1)u_0 \sum_{j < 0} h u_j m_a(u_j)$. By (H2) this is at most $(r-1)c_+ u_{l_0}^{1+\beta_a} \sum_{j \geq 1} h e^{-j\beta_a h} \leq (r-1)c_+ u_{l_0}^{1+\beta_a}/\beta_a$. Divided by the lower bound on I , it gives the second term of (27). It vanishes at $r = 1$.

Upper tail. The nodes $j \geq K$ are replaced by the delay mode, which is exact at $r = 1$. For $r \geq 2$ the residual is $\sum_{j \geq K} h u_j m_a(u_j) e^{-(r-1)u_j} \leq C_\infty e^{-(r-2)u_{hi}} \sum_{j \geq K} h u_j e^{-(1+\gamma)u_j}$ by (H3). The function $x \mapsto e^x e^{-(1+\gamma)e^x}$ decreases for $e^x \geq 1/(1+\gamma)$, so for $u_{hi} \geq 1$ the sum is at most $\int_{u_{hi}}^\infty e^{-(1+\gamma)u} du = e^{-(1+\gamma)u_{hi}}/(1+\gamma)$. Relative to I , this leaves the factor $(r-1)^{\beta_a} e^{-(r-2)u_{hi}}$. That factor is 1 at $r = 2$ and does not increase for $u_{hi} \geq \beta_+ \ln 2$, which gives the third term.

For (c), the choices of h , u_{l_0} and u_{hi} make the three terms at most $\varepsilon/2$, $\varepsilon/4$ and $\varepsilon/4$; the second uses $(R-1)u_{l_0} \leq 1$, so that the exponent $1 + \beta_a$ may be replaced by $1 + \beta_-$. A grid with $K = \lceil \ln(u_{hi}/u_{l_0})/h \rceil + 1$ nodes has a spacing at most h , and none of the three bounds increases when the spacing decreases. Finally, $1/h = O(\ln(1/\varepsilon))$ and $\ln(u_{hi}/u_{l_0}) = O(\ln R + \ln(1/\varepsilon))$, which gives (28). $\square$

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